

Cellulose-based materials for air purification: A review

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ABSTRACT

Background: Air-purification utilizing an air filtration system is vital in maintaining optimum interior air quality, despite the highly polluted air that anthropogenic activities have exacerbated. Since the constituents of air pollutants are so unstable and diverse, conventional air-filtering materials made of glass fibres or non-biodegradable polymers are facing a considerable dilemma in completely removing particulate pollutants of finer size like PM_{2.5} and gaseous pollutants like harmful chemicals, CO₂, and other contaminants. These synthetic polymers are likewise non-biodegradable, resulting in their disposal in landfills. Therefore, developing a fully biodegradable filtering system for air purification is undeniably urgent.

Scope and approach: Cellulose is a widely dispersed natural biopolymer and due to its porous, sustainable and surface-modifiable nature, it has demonstrated extraordinary relevance in numerous fields, including as high-performance air filtration system. In the present review, we discussed the significance, characteristics, functioning and ability of pure and surface functionalized cellulose and its derivatives-based filter system to remove particulate matter, volatile organic compounds and CO₂. In addition, different composite filters/membranes made using cellulose and its derivatives have also been discussed. Furthermore, we have also tried to investigate the mechanism for the absorption or removal of different pollutants utilising cellulose-based air filters. A comparative view of the properties of different cellulosic forms is also given here.

Key findings and conclusion: Cellulose is one of the essential polymeric materials in the air purifying industries. This study analyses the impact of various surface functionalisation techniques and blending materials such as carbon nanotubes, a natural biopolymer, polymer matrix, etc. on the performance of resulted filtering system. This review provides data pertinent to the use of cellulose in air purification based on an exhaustive analysis of the available literature. In addition, several future considerations are also discussed.

1. Introduction

Air pollution is called any physical, chemical or biological factor that contaminates the outdoor or indoor atmosphere and disrupts the environment's qualities. Domestic burning of appliances, industrial or factory operations, fuel burning, and wildfires usually causes it. Particulate matters (PMs), ozone, CO, NO₂, CO₂ and SO₂ are all severe public health concerns (Sheoran et al., 2022). Air pollution is a leading cause of respiratory and other ailments and key contributor to mortality rates. Every

year, approximately 7 million people die from air pollution worldwide. About 95% of the world's population lives in areas where air quality exceeds WHO standards (Anon, 2021). Even though both developing and developed countries are affected by ambient air pollution, middle- and low-income countries bear the brunt of the load. Both indoor and outdoor pollutants have a significant impact on the health of human beings (EPA, 2021; Anon, 2021).

Different ambient air polluting agents includes industry, automobiles exhaust, residential energy for cooking and heating, power generation

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and bio-waste combustion. These outdoor pollutants account for an estimated 4.2 million deaths yearly due to stroke, heart disease, lung cancer, acute and chronic respiratory disorders. Further, indoor air pollutants like cooking fire smoke are causing 3.8 million annual deaths per year (Anon, 2021). It can result in various health problems in both children and adults, ranging from respiratory ailments to cancer to vision impairments (Anon, 2016). When wood, coal and dung are burned in open hearths or inefficient burners, various types of pollutants such PM, CO, CO₂, CH₄, volatile organic compounds (VOCs; HCHO, phenol, etc.) and polyaromatic hydrocarbons (PAHs) are produced. These PMs are a hazardous contaminants (Dominici et al., 2014; Fan et al., 2018; Pope Iii et al., 2002). Numerous researchers have found a strong correlation between PM exposure and harmful health effects. Ultrafine particles ($\leq$ one-micron diameter) can permeate organs and tissues, presenting a much more significant risk of significant health effects. In addition to PM, formaldehyde, a crucial precursor of several chemicals and substances, has been a second significant source of indoor air pollution (Zhao et al., 2019). Indoor formaldehyde levels generally lie 10–100 times higher than outside levels.

PM's chemical constitutions can be highly complex, although it typically contains nitrates (NO₃), sulphate (SO₄²⁻ and SO₃²⁻), black carbon, mineral grits, and other substances (Bem et al., 2003). It may also combine with inorganic/organic particles present in the air to generate mixes/composites. Based upon the size of PM, these particles may be classified into different levels; for example, if a PM has an aerodynamic diameter $\leq 2.5 \mu\text{m}$ or $\leq 10 \mu\text{m}$, it can be referred to as PM_{2.5} and PM₁₀, respectively (Souzandeh et al., 2017a). Furthermore, PM_{2.5}, because of its small size, can pass easily into the human's lungs and bronchi, as well as blood vessels and can also act as a carrier for bacteria and viruses, aiding in the spread of respiratory diseases (Liu et al., 2020; R. F. Zhang et al., 2018; R. Zhang et al., 2018; Y.-G. Zhang et al., 2018; X.-F. Zhang et al., 2018; Zhao et al., 2017, p. 5). Every year, millions of people in China die due to PM pollution (Brook et al., 2010; Wu et al., 2014). Therefore, a high-efficient air filter should be utilised to trap PM of all sizes.

In addition to PM, increment in atmospheric CO₂ concentration has also attracted significant attention worldwide because it is a major reason for global warming and shares about 75% of total greenhouse gas emissions (Ansaloni et al., 2017; Dassanayake et al., 2016). One of the studies has recently claimed that after industrial revolutions CO₂ concentration in atmospheric air increased from 270 ppm to 400 ppm (Darunte et al., 2016). Therefore, researchers have concentrated on various technologies and devices to capture CO₂ or PM.

Traditionally, various fossil chemicals such as polyimide, polyacrylonitrile, polyethylene, polypropylene and glass fibres have been utilised as raw materials for fabricating commercial air filters. However, due to their size-based mechanism i.e. diffusion, interception and sieving, lack of active functional groups, higher cost, non-biodegradable, and non-recyclable nature, these materials face challenges for effective utilisation in preparation of air filters (Gough et al., 2021; Souzandeh et al., 2019).

Biopolymers such as silk fibroin, soy proteins isolate, keratin, cellulose and chitin, due to their easy availability, biodegradable, recyclable and straight forward to process nature have served as an alternative for preparing air filters, which in addition to superior filtration efficiency, also potentially address the issues mentioned above (Gough et al., 2021). Different filtration techniques have been used for air filtration making use of such polymers. We have recently summarised the challenges in air pollutants removal using biofiltration technique (Sheoran et al., 2022). Among different biopolymers, cellulose is one of the most abundant bio-polymers (Pappu et al., 2019; Platnieks et al., 2021). It has been utilised in variety of applications, including the manufacture of automobiles parts; bio composites (Rana and Singha, 2015), removal of heavy metals and pesticides from water, water desalination (Ashvinder K. Rana et al., 2021a; Rana et al., 2022b, 2022a), air purification (Gough et al., 2021), for air filtration (in vehicle

engine) (Dziubak and Dziubak, 2020), oil filtration (Feng et al., 2014) and fuel separation (Wang et al., 2021) in automobile industry (Rana, 2022), food packaging, as transparent membrane for ballistics protection (Sharma et al., 2019), biomedical applications (Picheth et al., 2017), etc. Thus, keeping in view its importance, we are in the present review article going to review its importance as air filter for eliminating PM, CO₂ and other VOCs from the atmosphere.

As per our knowledge, this is the first review where readers can go through a brief overview of the study and explain cellulose and filtration principles for removing PM and VOCs and CO₂ separation or adsorption. Furthermore, we have discussed the role of cellulose-based filters in air purification and CO₂ separation/adsorption (such as pure and hydrophobic/hydrophilic functionalized nano cellulose-based filter system, nanocellulose/other biopolymers-based filter systems, cellulose/carbon nanotubes-based, nanocellulose/metal-organic framework-based, cellulose/ionic liquids composites, carbonized cellulose-based, cellulose/polymer matrix-based, cellulose/polyoxometalates based, cellulose/metal oxide-based filter system, cellulose-based photonic films for detection of pollutants, and triboelectric based removal of particulate materials). Furthermore, the current challenges and future prospects of cellulose-based materials for air purification and environmental-related applications and finally concluded remarks of the recent study had also been discussed.

2. Cellulose and its derivatives

Cellulose is the one of the most abundant biodegradable polymers (polysaccharides) available in nature. It is a linear polysaccharide made up of β 1 $\rightarrow$ 4 glycosidic linkages between different D-glucose units and contains significant amounts of hydroxyl groups, primarily responsible for their highly rigid structure (Ashvinder Kumar Rana et al., 2021; Souzandeh et al., 2019; Ates et al., 2020). Further, with the help of hydrogen bonding they arrange themselves into parallel microfibril structure.

It gives physical strength to plant cell wall and its elementary nanofibrous structures provide high mechanical toughness and flexibility, in different plant species. Cellulose can be extracted from numerous sources such as annual plants, wood, micro-organisms, agricultural waste, and some animals (Zielińska et al., 2021; Beluns et al., 2021; Uppal et al., 2022). Nowadays, the synthesis, surface characteristics elucidation, and utilization of different forms of cellulose, i.e. cellulose nano crystals (CNCs) (Ashvinder Kumar Rana et al., 2021), cellulose nano fibrils (CNFs) (Souzandeh et al., 2019) and its derivatives such as Cellulose acetate (CA) (Y. T. Liu et al., 2021; Y. Liu et al., 2021), cellulose triacetate (CTA), ethyl cellulose (EC) (Y. T. Liu et al., 2021; Y. Liu et al., 2021), etc in numerous fields more specifically in the fabrication of air filters are gaining whose amount of attention among researchers all around the world (Carpenter et al., 2015; Y. T. Liu et al., 2021; Y. Liu et al., 2021; Nielsen et al., 2013; Siró and Plackett, 2010; Wei et al., 2014). The synthesis process, mechanical properties, density and dimensions of some of cellulose forms and their derivatives, which have most preferably utilized in fabrication of filters for removal of PM, CO₂ and VOCs from air have been displayed in Table 1 (Y. T. Liu et al., 2021; Y. Liu et al., 2021; McKeen, 2012; Ashvinder K. Rana et al., 2021b; Tristantini and Yunan, 2018; Volova et al., 2018; Wypych, 2016a, 2016b; Yang et al., 2013). Due to the highly crystalline structures, cellulose-based filters possess a high degree of thermal and chemical stability; yet, their rigid structure makes processing challenging (Delmer, 1999). Breaking it down for processing usually requires chemical treatment such as oxidation. When working with cellulose-based air filters, finding the correct combination of stiffness and process ability is crucial.

The hydrophilic or hydrophobic nature of cellulose and its derivatives play a significant role on the performance of the resulting filters in highly humidified conditions. Generally, hydrophobic cellulose derivatives or surface-functionalized cellulose filters usually are

Table 1

Types, synthesis and physic-chemical properties of some of the cellulosic forms and its derivatives.

Type of Cellulose or cellulose derivatives	Synthesis or extraction technique	Dimension/mechanical properties/density	Application	Degree of substitution, if any	Hydrophilic/hydrophobic nature	Reference
CA	- First time synthesized by Paul Schutzenberger from the pulp of wood using acetic anhydride - Procedure involves acetylation of cellulose using acetic acid in first step, which was subsequently dehydrated using sulphuric acid and finally treated with acetic anhydride to give CTA. CTA upon further acidic hydrolysis yields CA.	Micron sized Density: 0.52–0.73 g/cm ³ Tensile Strength: 40–41 MPa approx.	Used for preparation of membrane for bio-medical, air filters and other uses.	Extent of acetyl substitution varies from 29% to 48%. Degree of substitution: approx 1.5	Lower water swelling (0.28–0.65) and lower water content (25–36%), Less hygroscopic by nature	(Y.T. Liu et al., 2021; Tristantini and Yunan, 2018; Yang et al., 2013)
CTA	See cellulose acetate	Density: 1.28–1.34 g/cm ³ ; Tensile Strength: 28–56 MPa	See cellulose acetate	Degree of substitution: 0.89–2.84; Acetyl content: 60.7–61.2% (combined)	Water Adsorption: 2–7%	(Wypych, 2016a)
Ethyl Cellulose	Synthesised by reacting alkali cellulose with ethyl chloride.	Tensile strength of 76 μ m thick film: 47–72 MPa; Crystallinity: nearer to zero Density: 1.07–1.18 g/cm ³	Thin film coating material, vitamin and medical pills, binder and filler or as drug carrier	Extent of ethoxy substitution varies from 47.5% to 49%. Degree of substitution: 2.5.	Hydrophobic and insoluble in water. Soluble in organic solvents like ethanol, methanol, toluene, etc.	(Y.T. Liu et al., 2021; Y. Liu et al., 2021; McKeen, 2012; Wypych, 2016b)
CMC	Synthesised by reacting sodium chloroacetate with alkali cellulose solution.	Density (white powder): 0.5–0.7 g/cm ³ ; Crystallinity: 25.56%	Generally utilised in beverages	Degree of substitution: 0.38–1.4	Water loving	(El-Sakhawy et al., 2018; Holtzaple, 2003)
CNCs	Synthesised by treating the cellulose with ionic liquids (ILs), acids, solid acids, deep eutectic solvents, etc	Length: 100–500 nm; Diameter: 2–20 nm; Crystallinity: 54–88%; Tensile strength: 7500–7700 MPa Density: 1.6 g/cm ³	As fillers for fabrication of nano composites in diverse fields such as for fabrication of air filter, automobile parts, adsorbents, etc.	Nil (may have a degree of substitution if hydrolysed by sulphuric acid, oxalic acid, hydrochloric acid, etc.)	Highly hydrophilic	(Ashvinder Kumar Rana et al., 2021)
CNFs	Mechanical techniques are usually employed to extract CNFs from cellulose. However, pre-treatments such as 2,2,6,6-tetramethylpiperidine-1-oxyl radical (TEMPO)-mediated, enzymatic hydrolysis, etc. may be used in combination with mechanical method to minimize the energy requirement during mechanical destruction.	Length: 500–2000 nm; Diameter: 1–100 nm; Crystallinity: 45–80% Tensile strength: 357.5 MPa Density: 1.5 g/cm ³	As fillers for fabrication of nano composites in diverse fields such as for fabrication of air filter, automobile parts, adsorbents, etc.	—	Hydrophilic	(Ashvinder Kumar Rana et al., 2021)
BC	Produced by cultivation of specific strain in HS culture media consisting of monosaccharides or disaccharides.	Diameter: 25–80 nm; Length: millimetre Crystallinity: 45–89%; Young's Modulus: 3.4–5.5 MPa Density: 0.15 $\pm$ 0.01 cm ³	As fillers for fabrication of nano composites in diverse fields such as for fabrication of air filter, automobile parts, adsorbents, etc.	—	Highly hydrophilic	(Volova et al., 2018)
Cellulosic fibers	Extracted from plant by employing either water retting process or chemical process	Density: 1.2–1.5 g/cm ³ ; Percent crystallinity: 44–78%; Tensile strength (50 mm length): 147–570 MPa	May be utilized as adsorbents for heavy metals, pesticides, dyes from waste water. Also can be utilized to reinforce matrix for multidimensional uses	—	Least hydrophilic than CNCs, CNFs and BC	(Ashvinder K. Rana et al., 2021c)

preferred for better filtration at higher humidity.

3. Filtration principles

3.1. Removal of PM and VOCs

Air filtration utilizing cellulose-based air filters can be divided into two categories, i.e. physio-sorption and chemisorptions. In former one, harmful pollutants bind to filter surface or ingrained atoms/molecules

on the filtration system (Gough et al., 2021). Whereas in the later one i.e. chemisorptions, pollutants react with the filter membrane surface or ingrained molecules/atoms via both a non-catalytic or catalytic chemical way and forms simpler compounds.

The availability of the active/binding sites within the filtration system is the only limiting factor in the sorption technique. If no binding sites are available for the upcoming pollutants to bind onto the surface of filter, then remainder pollutants may flow across the filter. Just like traditional synthetic polymers based air filters, cellulose-based air filter

remove PM via impaction, interception, diffusion, and electrostatic interactions pathways, as depicted in Fig. 1 (Anon, 2003). The nature and size of pollutants strongly impact the primary filtration process. Pollutants having diameter > 1000 nm (spores and dust) are physically blocked by the filter, however smaller particles (100–1000 nm) are largely filtered via gravitation, straining, electrostatics interactions and fiber–pollutants interactions. Primary filtration process is quite effective in removal of PM.

In the case of nano fibers, interception and diffusion are more common than straining, gravitation and inertial impaction (Kadam et al., 2018; Sinclair, 1976). Interception occurs when fine PM (ideally $0.5 \mu\text{m}$) travel with the air stream and maintains a distance $\leq$ than their radius from the fibre surface. Diffusion, on the other hand, works for much finer particulates with aerodynamic diameters < 100 nm. For such fine particles, Brownian motion dominates and grabbing takes place through irregular collision with fibers' surface (Sinclair, 1976). Straining occurs mostly in Ultra-low particulate air filters (ULPA) and high efficiency particulate air (HEPA) filters, when PM flowing in the pathways between a couple or more fibers possess size less than the PM aerodynamic diameter (Yit et al., 2020). PMs having aerodynamic diameters in between 50 and 500 nm are oftenly eliminated by nanofilter system via Brownian diffusion and interception (Mukhopadhyay, 2010). In fact, the efficiency of nanofibers strongly relies on the particle's Peclet and Knudsen numbers, as well as the fibre nanofiber packing density and diameter. Fig. 1b also depicted the fractional collection efficiency for various mechanical filtration systems concerning the pollutant's diameter.

In electrostatic filtration, the charged functional groups implanted onto cellulose-based filter interact with charged pollutants and trap them. It is the most effective technique for removal of particles having diameter between 100–1000 nm (Gough et al., 2021; Anon, 2003). Further, the path of the pollutants through the filter can also be altered, which makes physical filtering easier. The electrostatically strengthened filtration system can also be used anywhere that contains a lot of ions, such as construction zones. Since, with times as charged particles start accumulates in the filter media, a decrement in electrostatic interactions between charged aerosols and the filtering material may be observed. Further, pressure drop in electrostatic type of filters increases at a very slow rate, so it might be not a good indicator to assess their performance or time of filter change. That's why, these kinds of filter should be checked and tested over the time for its complete discharging, for the purpose of its maintenance (change out time).

For the removal of VOCs or gaseous molecules, both physio-sorption and chemo-sorption work. In addition to electrostatic interaction,

hydrogen bonding, van der Waals interaction, hydrophobic interaction and polar-polar interaction also occurs during physio-sorption of pollutants onto cellulose bio-polymers based filters (Gough et al., 2021). Filters are typically built in layers to improve their pollutants removing efficacy (Liu et al., 2019). The maintaining low-pressure drop (ΔP), which is a measure of pressure change before (P_x) and after (P_y) filtration through the filter, ensures that the face mask or filters perform well. Low ΔP indicates good fluid/airflow through the filter mask/membrane and provides good breathability in the case of a face mask. When discussing porosity, ΔP , chemical functionalisation technique employed, blending of fillers, and layers filters, it's important to pay attention to the finer points, such as how the functionalisation procedure or the blending and geometry of stacking fibres affect filtration efficiency and wearer comfort. In general, as the thickness or number of layers in a filtration system or mask grows, the air permeability of the system or mask decreases. However, significant enhancement in the prevention of viruses and moisture has been noted after increased stacking of filters (Park and Jayaraman, 2021).

Even though filters have the same thickness or number of layers, different structures can be constructed between them, affecting filtration performance, ΔP , and air permeability. Another element that affects the mechanical and filtering performance of a filter is its porosity, the blending may easily regulate with suitable fillers, functionalisation, layering of fibres and geometry. Further, dust holding capacity, which simply tells us the amount of dust held per unit area of filter when pressure drop to a specified value, is also an important parameter and should be fully considered while developing an efficient filter media (H. Q. Zhang et al., 2019; H. Zhang et al., 2019). Generally larger the value of dust holding capacity of filter better will be its life span and vice versa. The notable feature for user compliance is the reusability of filters or masks thus special attention should be paid towards the recycling of filters and their performance. Further, types of filter fabrication processes may also impact their filtration performance, i.e. mechanical strength and porosity of filters, which has been discussed in the text below.

3.2. CO_2 separation or adsorption

The porosity of a cellulose-based membrane, which may vary with relative humidity, type of fillers used and surface functionalised techniques employed, determines the overall CO_2 permeability of the filter. In general, at a medium relative humidity (60–80%) or if fillers of suitable amount and porosity are used, the size of nanopores between nano-cellulose molecules increases due to swelling in the presence of

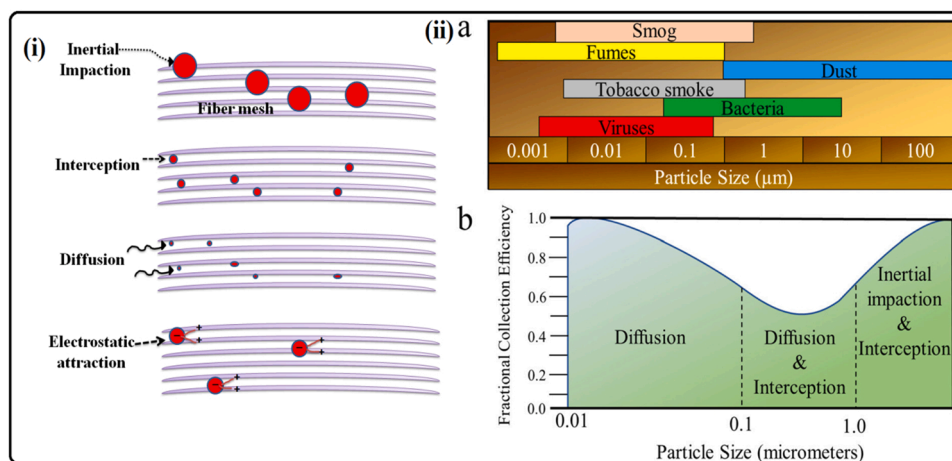


Fig. 1. (i) Typical mechanism showing the adsorption of PM or VOCs using fibres mash and (ii) (a) Relative size of common air pollutants and (b) fractional removal efficacy for different mechanical filters concerning the diameter of the pollutant (Gough et al., 2021; Anon, 2003). (Adapted from Ref (Gough et al., 2021; Anon, 2003) Copyright 2021, ACS).

moisture or due to interfacial adhesion between fillers and nano cellulose, to the point where CO_2 molecules, as opposed to CH_4 and H_2 molecules, can quickly diffuse through the nano-cellulose-based molecular sieve. The preference diffusion of CO_2 over CH_4 and H_2 may be due to the size differences between gaseous molecules ($\text{CH}_4 \sim 0.38$ nm, $\text{CO}_2 \sim 0.34$ nm, and $\text{N}_2 \sim 0.36$ nm). Since CO_2 has a small size thus, it may penetrate easily through the sieve of suitable pore size. Further, an increase in RH or fillers amount causes the polymeric matrix to swell more or form more voids in molecular sieve, resulting in larger space between nano-cellulose fibres (Dai et al., 2019a; Venturi et al., 2018). Thus, the pore size or gaps become wide enough for all gas molecules to pass through, causing the nanocellulose-based membrane to lose its selectivity (Fig. 2). As a result, pore size and humidity significantly influence gas separation.

In addition to separating CO_2 , it can be readily adsorbed onto nano-cellulose-based membranes. The deriving force behind adsorption is the interaction between CO_2 and functional groups on functionalised nano-cellulose-based membranes (Hafeez et al., 2015). Its adsorption is further influenced by the specific surface area, textural characteristics and densities of filters (Marco-Lozar et al., 2012; Zhuo et al., 2016). We'll also go through the numerous surface functionalisation techniques that have been used to adsorb CO_2 molecules in depth.

4. Role of cellulose-based filters in air purification and CO_2 separation/adsorption

Numerous cellulose-based filters /membranes have been developed to remove PM and adsorption/separation of CO_2 and other VOCs (Fig. 3). The details of which have been discussed below.

4.1. Pure nano cellulose-based filter

Freeze drying, electrostatic spinning, and heat drying are the most commonly utilised technologies for preparing CNFs or BC-based air filters (Golpira et al., 2021; T. Liu et al., 2021; Y. Liu et al., 2021). Each of these aforementioned technologies has its own importance and drawbacks. However, among them, electrospinning and freeze-drying are most suitable as no heating system is required in these technology (T. Liu et al., 2021; Y. Liu et al., 2021). Further, between freeze drying and electrospinning the later one was promised to be better in terms of higher parent compound preserving nature (Golpira et al., 2021). TEMPO-oxidized CNFs (TOCNFs) were dispersed in tert-butyl alcohol (TBA)/water mixtures and subsequently freeze-dried to produce to high-performance TOCNFs aero gel-based filters (Nemoto et al., 2015). When the TBA concentration was in the range of 20–50% w/w, the TOCNFs demonstrated homogenous dispersion in the TBA/water mixtures and higher specific surface area ($> 300 \text{ m}^2/\text{g}$). Furthermore, TOCNFs dispersion (30 wt%) in TBA/water mixture was combined with commercially available high-efficiency particulate air borosilicate micro glass fibre (BSMFs) and freeze dried to provide a spiderweb-like woven filter with exceptional filtration capabilities (99.999%) (Table 2).

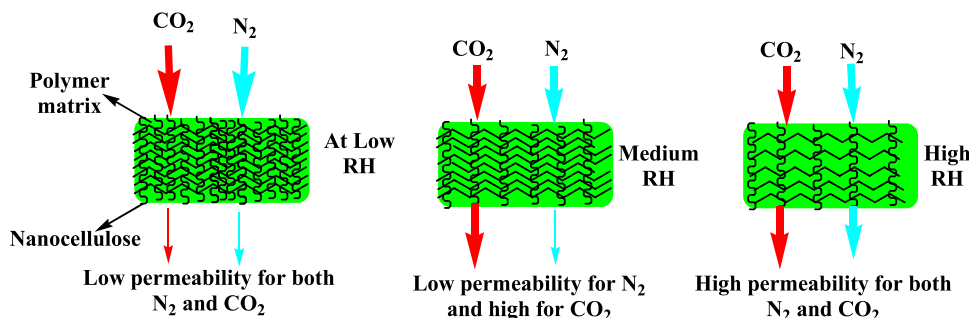


Fig. 2. Gas transport in nano-cellulose based membranes (Dai et al., 2019b).

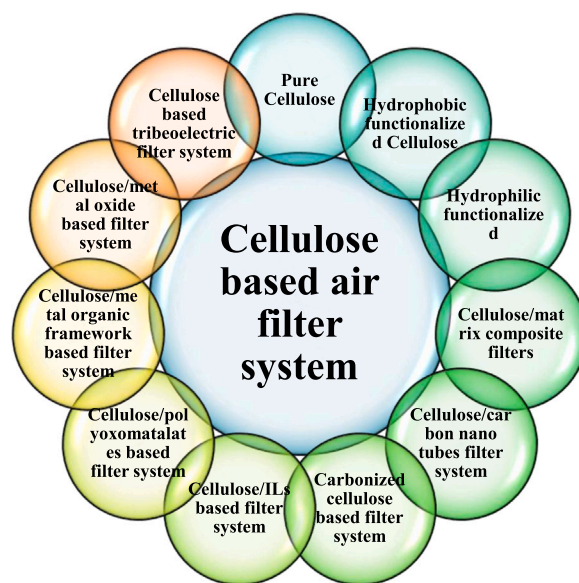


Fig. 3. Different cellulose-based filter systems have been developed and employed to purify the indoor/outdoor air.

The fibrillated cellulose fibres (CFs)/water mixture was freeze dried into a spiderweb-like structured air filter using TBA (Lu et al., 2018) (Table 2). The designed air filter showed a remarkable tendency to absorb (99%) $\text{PM}_{0.3}$. The TBA promotes separation between the fibrils CFs, transforming a lamellar porous network into a spider-web-like structure by altering the topologies and growth kinetics of ice crystals. The amount of TBA (varied from 10 wt to 40 wt%) played a significant role in enhancement in specific areas (increased from 143% to 218%) and air filter performance areas. The TBA contents at concentration greater than 30 wt% offered a $> 99\%$ removal efficiency for developed air filters.

In addition to TBA, organic solvents such as ethanol and iso-propanol were also used to synthesise cellulose-based filters using a freeze-drying method and then evaluated for their capacity to remove $\text{PM}_{0.3}$ and $\text{PM}_{0.5}$ (Ma et al., 2018b) (Table 2). The cellulose filters made up from water/10% TBA freezing media exhibited maximum porosity (96.4%), specific surface area ($3.8 \text{ m}^2/\text{g}$), and PMs removal efficiency (94.4% against $\text{PM}_{0.5}$), followed by filters made from water/10% isopropanol, water/10% ethanol, and pure water. Further, porosity, specific surface area, and removal efficiency increased to 98.2%, $12 \text{ m}^2/\text{g}$, and 99.7%, respectively, when the water/90% -TBA freezing medium was used. However, the pore size dropped from $120.9 \mu\text{m}$ to $60.5 \mu\text{m}$, $40.3 \mu\text{m}$, $27.8 \mu\text{m}$, and $4.5 \mu\text{m}$ with water/90%-TBA freezing medium and thus led to synthesis of spider web like structure. With 90% -TBA freezing media, the pore size dropped from 120.9 m to 60.5 m , 40.3 m , 27.8 m , and 4.5 m , resulting in synthesis of spider web structure.

Table 2

Data on PM removal efficiency of pure nanocellulose based filter system.

Filter	Surface functionalisation technique employed if any	Procedure Used	Adsorption efficiency	V (m/s)/ flow rate	Δ P	Ref
TOCNFs/ BSMFs	TEMPO oxidation	Freeze drying	PM _{0.125} :99.9996% PM _{0.175} :99.9998%	5.3	329	(Nemoto et al., 2015)
CNFs	—	Freeze drying	PM _{0.3} : > 99	5.3/ 32 L/min	380 approx.	(Lu et al., 2018)
CNFs prepared by water/ 10%-TBA freezing medium	—	Freeze drying	PM _{0.3} : 94.1% PM _{0.5} : 94.4%	5.3/32 L/min	105	(Ma et al., 2018b)
CNFs prepared by water/ 90%-TBA freezing medium	—	Freeze drying	PM _{0.3} : 99.66% PM _{0.5} : 99.7%	5.3/32 L/min	105	(Ma et al., 2018b)

4.2. Nanocellulose/other biopolymers-based filter systems

The PMs removal and CO₂ adsorption efficiency of various kinds of cellulose/biopolymers-based filter system has been depicted in Tables 3 and 4, respectively. Porosity and permeability of filter play a significant role in removing PMs (Table 3). Efforts were made to improve the permeability and porosity of bacterial cellulose (BC) by incorporating silver nanowires (AgNWs) into it (Wu et al., 2022). The synthesised (AgNWs/BC) filter offered a filtration efficiency of 99.7% and 99.8% when tested against PM_{2.5} and PM₁₀, respectively. The results obtained were comparable with the 3 M flask commonly utilised in the market. In addition to it, strong antibacterial activity was also noted for the composite material, which emphasises their potential to keep away human beings from germs and other bacterial infections. CNFs and Chitosan-based nano filters (CNFs/CS) also showed a remarkable ability to remove PM and CO₂ from the air (Sepahvand et al., 2021). The pore size and permeability of nano filters were found to be decreased with an increase in loading of CNFs; however, the opposite trend was observed for surface area. Highest PMs adsorption efficiency was found for CNFs/CS (2:1 wt%) nano filters (99.56%) succeeded by CNFs/CS filter (1.5:1) (97.93%), CNFs/CS (1:1 wt%) filter (95.72%) and pure CNFs (89.43%) against PM_{2.5}, respectively. Further, among different nano filters, CNFs/CS (2:1 wt%) nano filters displayed best particulate removal efficiency when tested against particulate materials having a sizes ranging from 0.1 – 2.5 μm. Further, adsorption ability of different

nano filters against carbon dioxide was found to be in the order: CNFs/CS (2:1) filter (4.8 mmol/g) > CNFs/CS (1.5:1) filter (3.5 mmol/g) > CNFs/CS (1:1) filter (2.6 mmol/g) > CNFs (2.2 mmol/g). The highest adsorption capability of CNFs/CS (2:1) filter has been linked to improvement in surface area upon an increase in % stacking of CNFs.

Many attempts have been made to construct a green multifunctional filter for PMs removal by combining soy protein isolate (SPI) with BC. SPI is the best of three types of soybean extracts: soy protein concentrates, SPI, and defatted soy flour, due to its high protein content (90%), low cost, and availability of a large number of functional groups onto its surface, which are generally exposed by denaturing with acrylic acid before blending with BC for better interaction with PMs (Liu et al., 2017). The BC 3D network assists in the initial trapping of PMs, while the functional groups of SPI further bind the PMs via electrostatic and dipole interactions between the adsorbents and the PMs. Adsorption efficiency of 99.95% and 99.4% against PM₁₀ and PM_{2.5} were achieved by combining BC with acrylic acid treated SPI (SPI/BC composite filter) in the right proportions. Further, the same group of researchers initially isolated the soybean proteins and separated them into 7 S and 11 S components utilizing Nagano method and subsequently used them to reinforce the BC filters (He et al., 2021). The SPI, 7 S and 11 S proteins were pre-treated with acrylic acid to increase the functional groups exposure and reduce protein size before being combined with BC to make three separate composite filters (PSPI/BC, P7S/BC and P11S/BC).

Table 3

Data on PM removal efficiency of nanocellulose/other biopolymers-based filter systems.

Name of Adsorbent	Type	Technique Employed	wt% of blended materials	PM removal ability	Airflow rate	ΔP (Pa)	Ref.
AgNW / BC	Membrane/ filter	In-situ cultivation using Staphylococcus aureus strain	28.57	PM _{2.5} 99.7% PM ₁₀ 99.8%	12 L/min	123 Pa	(Wu et al., 2022)
CNFs/Chitosan (2:1)	Filter paper	Emulsion freeze-drying technique	33 wt%	PM _{0.1} 92.52 PM _{0.3} 96.88 PM _{0.5} 97.77 PM _{2.5} 99.56	10 L/min	79 MPa	(Sepahvand et al., 2021)
4% acrylic acid -SPI/ BC	Composite films	Denaturation cum blending and refrigeration dryer	50 wt%	PM _{2.5} 99.94 PM ₁₀ 99.95	5.5 L/min	—	(Liu et al., 2017)
PSPI/BC P7S/BC P11S/BC	Composite films	Denaturation cum blending and refrigeration dryer	—	PM _{2.5} 73.07 PM _{2.5} 82.13 PM _{2.5} 85.44	5.5 L/min	—	(He et al., 2021)
CNs/corrugated paper	filter	TEMPO oxidation cum spraying	0.05% of CNFs sprayed in filter	PM _{3.0} 94.6	5.33 cm/s	174.2 Pa	(Xiong et al., 2021)
CNFs nano filters	Multilayered filter of thickness 330 μm	Electrospinning of CA followed by heating	0.14 g/cm ³ density	PM _{0.9} 98.4	3.0 L/min	167 Pa	(Souzandeh et al., 2016)
PDA-CNFs nano filters	Multilayered filter of thickness 330 μm	PDA coating onto CNFs filter	0.14 g/cm ³ density	PM _{0.9} 98.9	3.0 L/min	162 Pa	(Souzandeh et al., 2016)
Cellulose /CA	Multilayered filter consisting of 2 layers of cellulose and one layer of CA	—	—	PM _{0.9} 99.8	3.0 L/min	158 Pa	(Souzandeh et al., 2016)
Protein functionalized BC/microcellulose	Composite filter	—	—	PM _{1–2.5} 99.5%	4 cm/s	0.194	(Fan et al., 2018)
GNFs/PT/GNFs	Hybrid filter	Electrospinning	—	PM _{0.3} 99.2% PM _{0.3–10} 100	10 L/min	195 Pa	(Souzandeh et al., 2017b)

Table 4Data on adsorption of CO₂ and other VOCs from nanocellulose/other biopolymers-based filter systems.

Composite material	Type of foam	Synthesis approach	Testing conditions	Flow rate (L/min)	Uptake/permeability	Ref.
CNFs/Chitosan (2:1)	Filter paper	Emulsion freeze-drying technique	33 wt % 1 BAR pressure at room temperature; 30% relative humidity; 8 hr	0.8	4.48 mmol/g	(Sepahvand et al., 2021)
GNFs/PT/GNFs	Hybrid filter	Electrospinning	—	4	HCHO: 83.70 ± 1.10 CO: 80.22 ± 1.40 SO ₂ : 81.77 ± 0.46 VOCs: 83.70 ± 0.55%	(Souzandeh et al., 2017b)

P7S and P11S were found to have a uniform or non-aggregated coating on the BC surface; however, PSPI had the reverse effect. The adsorption order for PM_{2.5} was found to be: P11S/BC (85.44%) > P7S/BC (82.13%) > PSPI/BC (73.07%). A hierarchically structured air filter was constructed by spraying CNF suspensions of various concentrations (0.02–0.20%), sizes and masses (2.5–10 gm) over the corrugated paper (Xiong et al., 2021). Here, porous structures of corrugated paper

functioned as frame, while CNFs served as functional fillers. A circular corrugated paper with a diameter of 16 cm, with a base weight of 128 g/m² and a pressure of 30 Pa was utilized for spraying. After the spraying of CNFs onto paper, the air filter was freeze dried and subsequently utilized to remove PM_{0.3}. Removal efficiency of 94.6% was recorded when CNFs of 0.25 g/m² base weight was loaded in the air testing filter.

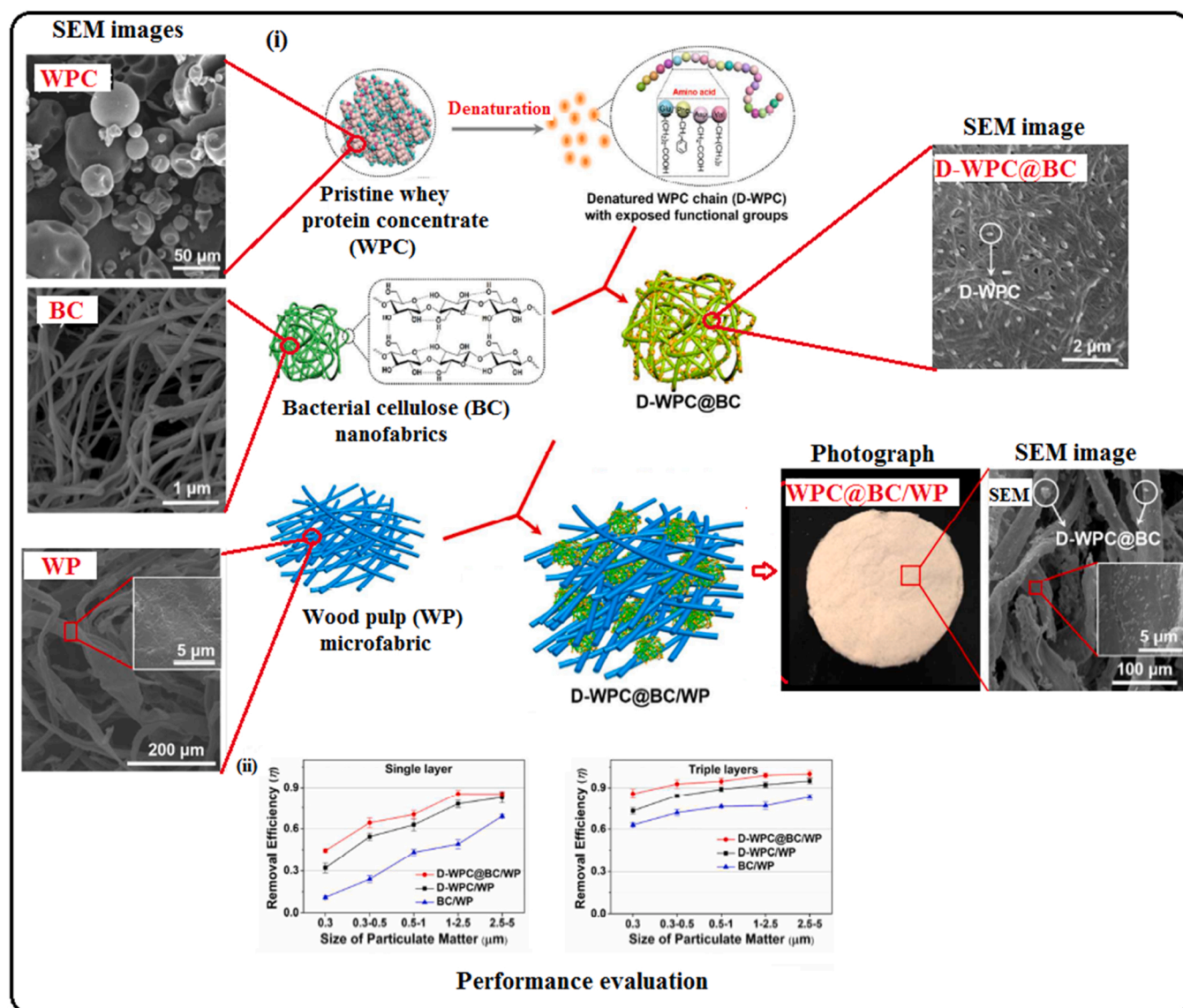


Fig. 4. (i) Preparation strategy and SEM images of different precursor materials and resulted WPC@BC/WP composite filters and (ii) their performance against different PMs (Fan et al., 2018). Reproduced with permission from Ref. Fan et al. (2018) Copyright 2018, ACS.

Cellulose multi-layered nanofillers (CMNFs) of thickness 330 μm (formed by combining three CNFs filters of each of thickness 110 μm), regenerated from electrospun CA and subsequent heating at 200 $^{\circ}\text{C}$, under highly hazardous environment (PM concentration $\sim 2000 \mu\text{g}/\text{m}^3$) offered ΔP , filtration efficiency and quality factor to be $< 200 \text{ Pa}$, $> 98.4\%$, and $\sim 0.03 \text{ Pa}^{-1}$, respectively (Souzandeh et al., 2016). A considerable increase in removal efficiency (99.8% at RH 80%) was observed when a hydrophobic layer of CA was stacked below the two layers of hydrophilic cellulose. The efficiency of CNFs based multilayered filters was noticed to enhanced from 98.4% to 98.9% after coating of bio-compatible hydrophilic polydopamine (PDA) on CNFs; however, when coated with hydrophobic polycation poly(diallyldimethylammonium chloride) (PDADMAC), filtration efficiency drops to 97.9%. The impact of packing density (varied from 0.10 to 0.18 g/cm^3) and filter thickness (110–330 μm) was also evaluated, and maximum efficiency was noticed at a density of 0.14 g/cm^3 and 330 μm thickness.

Protein-based nanofilters possess very high removal efficacy for toxic chemicals because of their vast surface area and abundance functional groups that can create strong interactions with various pollutants (Hung et al., 2021; Li et al., 2014). Recently, researchers attempted to functionalise BC with whey protein nanoparticles before combining it with wood pulp microcellulose to create a highly efficient composite filter that eliminated 99.5% of $\text{PM}_{1-2.5}$ particles (Fan et al., 2018) (Fig. 4). In addition to capturing pollutants, protein-functionalized nanocellulose also acts as a binder to reinforce the composite filter. Recently, gelatin nanofibres (GNFs) were coated onto both sides of the paper towel (PT) using facile electrospinning technique to develop hybrid filtering materials (GNFs/PT/GNFs) (Souzandeh et al., 2017b). The GNFs/PT/GNFs filtering material displayed approximately a removal efficiency of 100% for $\text{PM}_{0.3-10}$, whereas against $\text{PM}_{0.3}$ a removal efficiency of approx. 99.2% was observed. Aside from PM, the synthesised materials also showed great promise against VOCs, formaldehyde, carbon dioxide, and other harmful substances. An adsorbing efficacy of 83.70 ± 1.10 , 80.22 ± 1.40 , 81.77 ± 0.46 and $83.70 \pm 0.55\%$ was noted against HCHO, CO, SO_2 and VOCs.

4.3. Surface functionalised hydrophobic nanocellulose based filter/membrane system

Hydrophobically functionalized biopolymers with a water contact angle of more than 90° , have attracted a broad range of applications (Anjum et al., 2020). The amazingly low water adherence nature of these bio-polymers causes droplets of water to bead and thus reduces the degradation of their surface composition (Dou et al., 2015). Furthermore, due to their resistance to air humidity, these materials are being used in diverse fields such as the construction industry, photovoltaic energy, food packaging, etc.

Although, hydrophilic air purification filters can be degraded by high air humidity, efforts were made to hydrophobize these air filters and employ them to improve air quality (Agustín-Sáenz et al., 2020; T. Liu

et al., 2021; Y. Liu et al., 2021). The CNFs were super-hydrophobized (CA: 152.4 $^{\circ}\text{C}$) by methyltrimethoxysilane (MTMS) (Table 5) coupling agent (T. Liu et al., 2021; Y. Liu et al., 2021). The CNFs based aerogel filter showed a high adsorption effectiveness of 97.96% and 99.80% against $\text{PM}_{0.3}$ and $\text{PM}_{2.5}$, respectively. Further, the super hydrophobised CNFs successfully resisted the interference of air humidity or moisture on their performance and also offered an increase in water vapour transmission rate up to 1.24 times. Thus, MTMS functionalized CNFs filters have great potential as open air dust collectors.

Development of hydrophobic character onto bamboo cellulose by treating with (3-chloro-2-hydroxypropyl) trimethylammonium chloride (CHPTAC) followed by ion exchanging of pre-synthesised materials with sodium carbonate has been reported by Hou et al. (Hou et al., 2019, p. 2). The resulted quaternized cellulosic materials offered better performance at higher humidity levels (66%) for the removal of carbon dioxide when compared to hydrophilic ion exchange resin (IER). Further, at 65–70% humidity level, the quaternized bamboo cellulose showed an adsorption capability of 0.18 mmol/g for carbon dioxide. Further, polydimethylsiloxane (PDMS) treatment was applied to develop hydrophobic character and boost the performance or life of cellulosic filter, polydimethylsiloxane (PDMS) treatment was applied onto pre-synthesized porous CNFs/stainless steel based thin layer filter, which was created via simple filtration-supported gelation technique (Chen et al., 2017). The produced hydrophobic filter showed quick air permeability and long life span, even when employed at higher relative humidity, i.e. 49–93%. The hydrophobic CNFs/stainless steel screen offered a removal efficacy of 96% for $\text{PM}_{2.5}$. Furthermore, the air permeance of the hydrophobised CNFs/stainless steel screen was discovered to be substantially closer to the wind speed on days with heavy PM pollution in the range of 0.1–1.0 m/s. Recently, Lotus leaf was powdered into CNCs form using the ball milling technique, which was subsequently used to adsorb formaldehyde VOCs from indoor air (Yang et al., 2020). When ball milling periods were increased from 24 to 120 hr, a significant decrease in WCA ($147.5\text{--}45^{\circ}$) of prepared CNCs was observed. After 150 min of adsorption, CNCs prepared after 120 hr of ball milling were found to eliminate 98% formaldehyde.

Furthermore, to improve CO_2 adsorption efficacy, CNFs were chemically modified by utilising phthalimide (0.5, 1 and 1.5 wt%) and then freeze dried to obtain CNFs-phthalimide filters (Sepahvand et al., 2020). With the increase in phthalimide, the specific area of resulted filter was found to increase; however, the size and volume of pores and overall porosity decreased. In addition, variation in pressure, temperature, and relative humidity significantly impacted the CO_2 adsorption ability of CNFs-phthalimide filter, without disturbing the CNFs structure.

4.4. Nanocellulose functionalized with a hydrophilic polymer

Polyethylenimine (PEI) is a prospective alternative to the existing chemical agents for functionalizing cellulose-based sorbents for CO_2 uptake because of its highly hydrophilic nature (Kuwahara et al., 2012;

Table 5
Comparison of filter performance of hydrophobic cellulose-based filters.

Filter	Surface functionalisation technique employed	Procedure Used	Adsorption efficiency	V (m/s)/flow rate	Delta P	Ref
CNFs based aerogel	Silanized using MTMS coupling agent	Freeze drying	$\text{PM}_{0.3}$: 97.96 $\text{PM}_{2.5}$: 99.80 CO_2 : 0.18 mmol/g	5.3	104	(T.T. Liu et al., 2021; Y. Liu et al., 2021)
Bamboo cellulose	CHPTAC treatment	Grafting cum Ion exchange technique		–	–	(Hou et al., 2019, p. 2)
CNFs	Simple filtration assisted-gelatin process followed by Polydimethylsiloxane	vacuum-drying	$\text{PM}_{2.5}$: 96%	10	200	(Chen et al., 2017)
CNCs		Ball milling of lotus leaf for 120 hr	98% of formaldehyde adsorbed	–	–	(Yang et al., 2020)
CNFs-phthalimide (1:1.5 wt%)	Phthalimide treatment	Freeze drying	7.1 mmol/g at 8 hr time and 1 bar pressure	0.8 L/min	150	(Sepahvand et al., 2020)

Wang et al., 2012). TEMPO-CNFs/PEI sorbents were prepared by varying the PEI contents from 19.3 to 62.0 wt% and tested for their ability to absorb CO₂ under atmospheric conditions (0–80% Relative Humidity; 400 ppm of CO₂) (Sehaqui et al., 2015). At 80% RH, 44 wt% PEI loaded TEMPO-CNFs/PEI sorbents offered the highest adsorption capacity of 2.22 mmol/g. Furthermore, the prepared foam remained stable after five cycles and achieved an extremely short adsorption half time of 10.6 min

4.5. Cellulose/carbon nanotubes-based filter system

Carbon nano tubes (CNTs) are widely utilised in aerogel production due to their alluring properties such as low mass density, uniform pore diameter, high mechanical toughness, and high specific surface area. Since CNFs aerogel possesses lower surface area, so attempts were also made to enhance the surface area by incorporating CNTs (Gao et al., 2013; Wang et al., 2013). A nanostructured aerogel (deodorizer),

comprising of TEMPO oxidised CNFs, divalent transition metal cations (Cu, Ni and Co) and multi-wall carbon nanotubes (MWCNs), was developed utilising freeze-drying techniques (Adavan Kiliyankil et al., 2021). Here, CNFs and MWCNs play a backbone, whereas M²⁺ works as a linker. The resulting foam like aerogel (CNFs-M²⁺-MWCNTs) showed better capability than CNTs free CNFs-M²⁺ aerogels to selectively adsorb NH₃, H₂S, CH₄S, trimethylamine (C₃H₉N). Further, these materials maintained their adsorbing efficacy even after the tenth cycle, making them a potential candidate for industrial utility. Among different types of aerogels, CNFs-Cu²⁺-MWCNTs offered the highest removal efficiency and achieved 89% within 10 min and 99.3% in 30 min for NH₃ (150 ppm); 77% within 10 min and 97% in 30 min for trimethylamine (70 ppm), 100% removal against hydrogen sulphide (20 ppm) within 10 min and 100% absorbance for mercaptan (100 ppm) within 30 min. Mixed matrix membrane made up by mixing cellulose acetate and β-cyclodextrin functionalised MWCNTs via wet phase inversion technique showed a considerable potential for separation of CO₂/N₂ (Ahmad

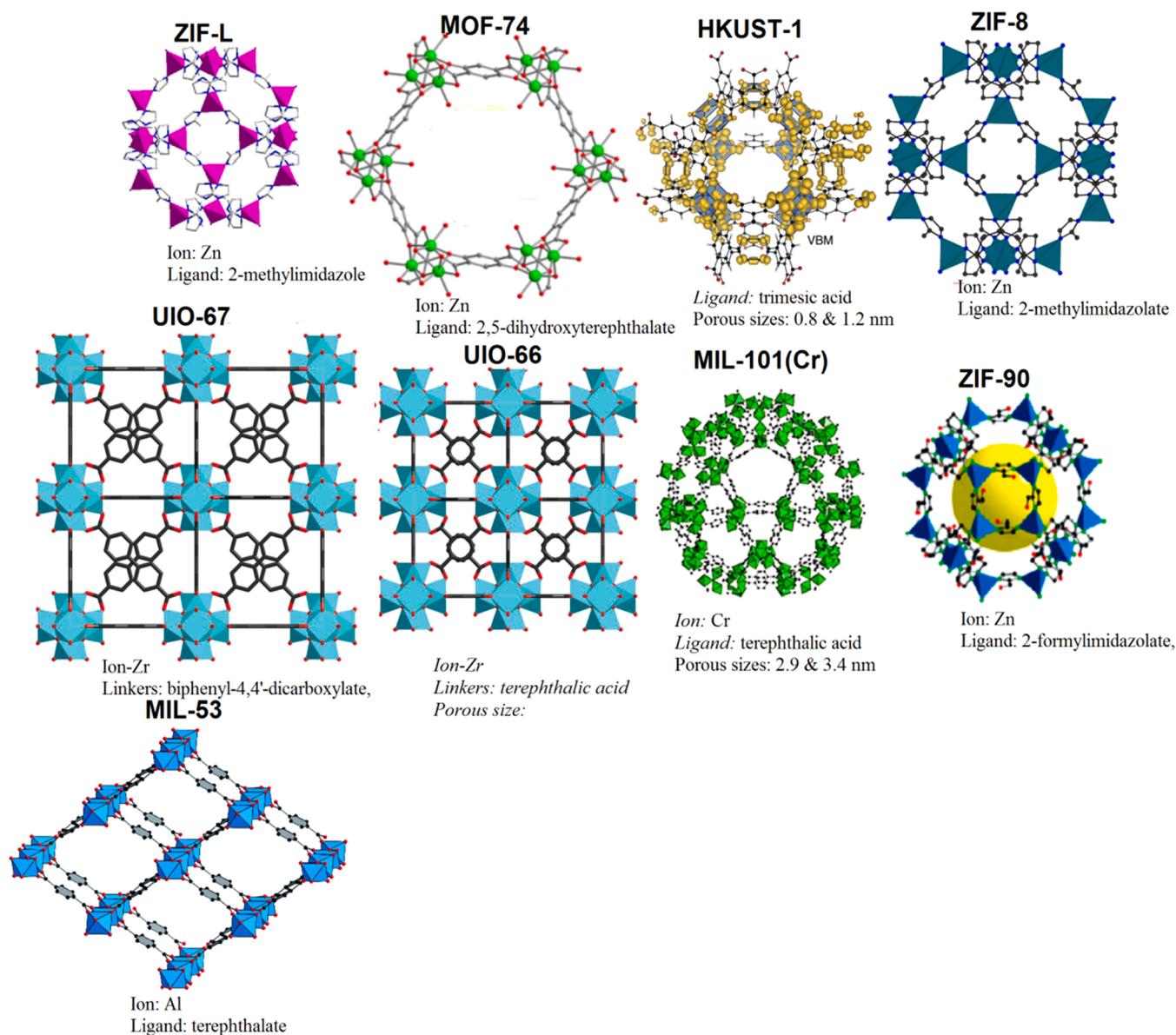


Fig. 5. Displaying the structure of some commonly utilised MOFs for synthesis of nanocellulose/MOFs filter system. [All images has been reprinted with permission from respective publishing house; MOF-74 image has been reprinted from Ref (L. Zhang et al., 2016; Q.G. Zhang et al., 2016), MIL-53 image from Ref (Nguyen et al., 2019) under Creative Commons Attribution-Non Commercial 3.0 Unported Licence, ZIF-L image from Ref (Li et al., 2017), UIO-66 & 77 from Ref. (Yuan et al., 2018), ZIF-8 & 90 from Ref (Phan et al., 2009), MIL-101 from Ref. (Trung et al., 2010) and HKUST-1 image from Ref. (Hendon and Walsh, 2015).

et al., 2014). The % loading of functionalised MWCNTs was varied from 0% to 0.2% however, the maximum separation ability of resulted membrane was discovered at 0.10 wt% loading of MWCTNs, where it demonstrated good CO₂/N₂ permeance rate (741.67 GPU: 84.98%) and selectivity up to 40.17. Thus, we can conclude that interfacial interaction between MWCNTs and cellulosic materials plays a crucial role in removal of gaseous as well as particulate materials.

4.6. Nanocellulose/metal organic framework-based filter system

Materials with high specific surface areas combined with an interconnected nonporous structure are frequently required for selectively adsorbing noxious gases from the air, which is practically unattainable from different pure forms of cellulose (S. Zhou et al., 2019a). As a result, various researchers have attempted to combine metal organic frameworks (MOFs), a new class of crystalline nanoporous materials, with cellulose substrates and use them to remove noxious gases from the air.

MOFs are 3D microporous inorganic-organic crystalline materials, which are formed by self-assembly of metals ions and organic linkers through co-ordinate bonds. MOFs offered high thermal strength, adequate mechanical strength, larger surface area from 200 to 7000 m²/g, low density below 0.13 g/cm³, uniform cavities of size 0.3–0.10 nm and high adsorbing ability (Chen et al., 2018; Deng et al., 2012; Furukawa et al., 2011). Due to crystalline, highly porous nature with large surfaces, controllable pore size, and functional groups and presence of defects and an asymmetry of surface metal ions on MOF frameworks, these materials has demonstrated impressive tendency to remove air impurities via electrostatic interaction (Lu et al., 2015; Yuan et al., 2016; Zhao et al., 2019). Fig. 5 displayed structure of some of the MOFs, along with nature of metal ions and ligands, which has been utilised for adsorption of air pollutants and removal of CO₂ from air.

Further, to enhance the adsorbing capability or air permeability across MOFs membrane, which is often limited due to their dense packing, its surface tailoring or blending with other porous materials is primarily required. Different types of porous polymeric materials can be mixed with MOFs, but cellulose fibres have several advantages, including being inexpensive, easy to process, and having a synergistic effect (Abdelhamid and Mathew, 2022).

These polymers, when combined with cellulosic materials such as CNFs or CNCs, have been proven to have significant potential to adsorb/remove various air pollutants such as formaldehyde, PM, and other contaminants (Bian et al., 2018). Combining a low-cost biodegradable

cellulose nanofibrous structure with multifunctional MOFs will not only save money, but will also result in the most effective removal of toxic gases from the air (Xu et al., 2018; L. Zhang et al., 2016; Q.G. Zhang et al., 2016; Zhou et al., 2019b).

The blending of Cellulose and MOFs can be carried out by utilising in-situ or ex-situ synthesis technique (Abdelhamid and Mathew, 2022). Each technique has its own advantages and disadvantages. The details of which have been discussed in Table 6. In in-situ synthesis technique, MOFs precursors (metals or linkers) are added to the cellulose-based materials. Whereas in case of ex-situ approach, pre-synthesised MOFs are added directly to cellulosic fibers or foams.

Cellulose-MOFs materials offers porous forms with improved flexibility and customizable hierarchical porous structures (Zhu et al., 2016).

4.6.1. Removal of particulate matters

MOFs/Cellulose composites have shown remarkable potential in the removal of PMs (Table 7). For adsorption of PM_{2.5}, a composite filter paper (ZIF-8/Chitosan/Cellulose) was made by (i) growing ZIF-8 crystals onto chitosan gel (CS) using an in-situ technique in first step and (ii) then subsequently by adding it with cellulose pulp (Nie et al., 2020). The filter paper showed 99.68% removal efficiency against PM_{2.5}. Furthermore, the designed filter paper offered excellent permeability and also sustain its efficacy over a wide range of aerodynamic size (10–0.3 μm) (Fig. 6). Zeolitic imidazolate framework-8/cellulose fibers (ZIF-8/CFs) filters were fabricated by Ma et al. by utilizing in-situ growth method (Ma et al., 2018a). A higher adsorption efficiency of about 99.94% and 98.36% were noted for the composite filter against 0.5 and 0.3 μm particulate. In addition, nitrogen adsorption tendency was also noticed to enhance by about 20 times compared with cellulosic fibres.

Attempts were also made to adsorb PMs (generated by Diethyl hexyl sebacate aerosol (DEHS) nebulizer and having 0.3 μm diameter) utilizing Cellulose/MOFs (ZIF-8/cellulose). ZIF-8/CFs showed an adsorbing ability of about 99.9% against PM_{0.3}. The synthesized filter was found to be as effective as the N99 industrial filter for the simultaneous removal of PM_{2.5} and other toxic gases such as HCHO, SO_x, CO, NO_x, etc (Su et al., 2018). Four different types of cellulose or MOFs/Cellulose based filters (cellulose filter (CF), Ag-MOFs@CNFs@ZIF-8, ZIF-8 @CNFs@CF and Ag-MOFs@CNFs@CF) were synthesised and utilized for the removal of PM_{2.5} and PM₁₀ (DEHS aerosol generated by nebulizer). Highest filtration efficiency (against both PM_{2.5} and PM₁₀) was found in case of Ag-MOFs@CNF@ZIF-8 (94.3% and approx. 91%, respectively) followed by ZIF-8 @CNF@CF, Ag-MOFs@CNF@CF and pure CF (Ma et al., 2019).

Table 6
Different approaches for doping of MOFs onto cellulose based materials (Abdelhamid and Mathew, 2022).

Approaches	Procedure	Methods utilised for synthesis of MOFs	Advantages	Disadvantages
In-situ approach	- MOFs precursors are added to cellulosic fibers or cellulose forms, aerogels, scaffolds, etc. The cellulosic fibers after MOFs doping may be processed into foams, aerogels or in other desired foams using conventional processing methods such as vacuum filtration, solution casting, solution blending, 3D printing, layers-by-layers, dip coating, electrospraying and doping methods. - MOFs precursors can be added directly to cellulose forms, aerogels, scaffolds, etc	Solvothermal or hydrothermal methods	- Can enhance the mechanical toughness of fibers - Leads to homogenous distributions - Simple procedure - Strong interaction is possible between cellulose fibers and MOFs - Loading can be increased by successive addition of MOFs precursors	- Control of MOFs loading into Cellulose fibers or cellulose foams is difficult. - The change in cellulose's conformational structure may occur because of deep diffusion of MOFs. - Prevent the diffusion of pollutants into MOFs crystals.
Ex situ approach	- The pre-synthesised MOFs are applied to cellulose fibers or cellulose forms to form mixed matrix membranes. - The three steps are the foundation of this method. MOFs are first synthesised using traditional methods. The pre-synthesised MOFs are then added to cellulose fibres or cellulose form in the second stage. In the final step, cellulose-MOFs are processed into the desired form using standard processing methods (as discussed in case of the in-situ approach).	Sol-gel procedure	- Easy to control loading of MOFs into cellulose. - Provides a large surface area and high adsorption capacity - High accessibility to reactants and chemical reagents	- Time consuming technique, - Chemical reagents are required - Severe aggregation of MOFs may occurs - Result in non- uniform distribution

Table 7

Removal of PM from the air using MOFs/Cellulose-based materials.

Type of Adsorbent	Synthesis Procedure	Form	% wt. loading of MOFs	Size of PMs	Removal Efficacy (%)	Flow rate	Reference
ZIF-8/Chitosan/Cellulose	In-situ synthesis	Paper	11.67	PM _{2.5}	99.68	32 L/min	(Nie et al., 2020)
ZIF-8/CFs	In-situ synthesis followed by freeze-drying	Filter	19.75 based on Zinc %	PM _{0.3} PM _{0.5}	98.36 99.94		(Ma et al., 2018a)
ZIF-8/CFs	In-situ synthesis followed by freeze-drying	Filter	90.2	PM _{0.3}	99.9	32 L/min	(Su et al., 2018)
Ag-MOFs@CNF@ZIF-8	In situ green deposition method	Filter	10 based on Zinc %	PM _{2.5} PM ₁₀	94.3 91% approx	32 L/min	(Ma et al., 2019)
HKUST/CNFs/stainless steel	filtration-assisted gelation process	Membrane	–	PM _{2.5}	95.0	4.6 cm/s	(Zhao et al., 2019)
UIO-66_30°C/cotton	In-situ synthesis	Fabric	20	PM _{2.5} PM ₁₀	65.0 68.0	0.2 m/s	(Yoo et al., 2020; Yoo and Jung, 2019)
UIO-66-NH ₂ /cotton	In-situ synthesis	Fabric	20	PM _{2.5} PM ₁₀	72.0 78.0	0.2 m/s	(Yoo and Jung, 2019)
UIO-66-NH-SO ₃ H/cotton	In-situ synthesis	Fabric	20	PM _{2.5} PM ₁₀	80.0 85.0	0.2 m/s	(Yoo and Jung, 2019)
UiO-66-NO ₂ /cotton	In-situ synthesis	Fabric	18.56	PM _{2.5} PM ₁₀	89.5 91.0	0.2 m/s	(Woo et al., 2020)
UiO-66-NH ₃ ⁺ Cl ⁻ /cotton	In-situ synthesis	Fabric	18.47	PM _{2.5} PM ₁₀	77.3 82.2	0.2 m/s	(Woo et al., 2020)
Cotton	–	Filter	18	PM _{2.5} PM ₁₀	36.9 46.5	11 L/min	(Yoo et al., 2020)
UIO-67_80°C/cotton	In-situ synthesis	Fabric	18	PM _{2.5} PM ₁₀	73.3 78.0	11 L/min	(Yoo et al., 2020)

Further, dual MOFs were found to be more effective than a filter with single MOFs.

UIO-66 MOFs with or without functional groups (SO₃H and NH₂) were coated onto conventional cotton to remove PM₁₀ and PM_{2.5} from air (Yoo and Jung, 2019). The PMs removal efficiency of resulted filters (UIO-66/cotton, UIO-66-NH₂/cotton and UIO-66-NH-SO₃H/cotton) was significantly enhanced after the introduction of functional groups onto MOFs. The highest adsorption efficacy, when loaded with 10 wt% of MOFs, was found for UIO-66-NH-SO₃H/cotton (70% in case of PM_{2.5} and 75% in case of PM₁₀) followed by UIO-66-NH₂/cotton (57% and 65%), UIO-66/cotton (55% and 60%) and cotton (31% for PM_{2.5} and 46% for PM₁₀). The increase in adsorption efficacy is due to electrostatic interaction between carbonyl groups onto the outer surface of PM and SO₃H groups onto MOFs. Further, with an increased percent loading of MOFs from 10 to 20 wt%, adsorption tendency was further noticed to enhance from 70% to 80%, 57–72% and 55–65% against PM_{2.5} and from 75% to 85%, 65–78% and 60–68% against PM₁₀ in case of UIO-66-NH-SO₃H/cotton, UIO-66-NH₂/cotton and UIO-66/cotton, respectively. The same group further extended their efforts and developed two more MOFs/-Cotton based filters having different surface functionalities (UiO-66-NH₃⁺Cl⁻/cotton and UiO-66-NO₂/cotton (Woo et al., 2020). The observed adsorption efficiency in this case was in the following sequence: UiO-66-NO₂/cotton (89.5% in case of PM_{2.5} and 91.0% in case of PM₁₀) > UiO-66-NH₃⁺Cl⁻/cotton (77.3% and 82.2%) > UiO-66-NH-SO₃H/cotton (73.8% and 77.7%) > UiO-66-NH₂/cotton (68.6% and 74.2%) > UiO-66/cotton (65.3% and 68.3%) > cotton (36.9 and 46.5 in case of PM_{2.5} and PM₁₀, respectively). The porosity of MOFs plays a significant role in the adsorption efficiency of resulted materials against PMs. Yoo and co-workers (Yoo et al., 2020) investigated the impact of different MOFs' porosity (DUT-52, UiO-66 and UiO-67) on the adsorption efficiency of resulting MOFs/Cotton filters against PM_{2.5} and PM₁₀. The adsorption efficiency was noted to be increased with an increase in porosity of MOFs, with the following trend: UiO-67/cotton > DUT-52/cotton > UiO-66/cotton.

Traditionally, electric motors were used to purify indoor air. Still, because of their high maintenance costs and high energy consumption, scientists are now focusing their research on bio-degradable and biocompatible cellulose/MOFs-based filter systems (Zhao et al., 2019). Porous CNFs/HKUST-1 membrane generated on a stainless screen was utilised as window screen for blocking PM_{2.5} and HCHO. The membrane

offered a blocking efficiency of 95% against PM_{2.5} for 24 h and an adsorption capability of 47.71 mg/g for HCHO (Zhao et al., 2019). In addition, the airflow rate (4.6 cm/s) demonstrated by CNFs/HKUST-1 membrane at 200 Pa was found to be sufficient for indoor activity. Further, light volume, low density and filtration-assisted gelation procedure will be favourable for their mass production on industrial scale, packaging, and mass transit/shipping. It may boost its global competitiveness in the field of air purification (Zhao et al., 2019).

4.6.2. Removal of VOCs and CO₂

A comparative view of different MOFs/cellulose-based filter system in adsorbing CO₂ has been given in Table 8. Zhou et al. assembled MOFs into continuous nanolayers of Cladophora Cellulose (CC), extracted from green algae, to form hybrid CC@MOF nanofibers to increase the effective surface area of resulted materials for the gas removal (S. Zhou et al., 2019b). Their procedure includes addition of a moderate amount of polyvinylpyrrolidone (PVP) into the suspension of ion-exchanged carboxylated CNFs to modify the surface of the CNFs, followed by drop wise inclusion of a certain proportion of organic ligands and metal salts in resulting suspension with continuous stirring. Four different types of MOFs (ZIFCO₃-1, HKUST-1, Zn-MOF-74, Al-MIL-53) were grown onto nano fibrillar layers by tuning dosage of PVP and the degree of carboxylation /charge density of CNFs, and finally each composite specimen was converted into flexible, mechanically more challenging, thermally stable, and highly transparent nano papers form (CNF@MOF nanopapers). The results materials have been reported to have a high VOCs separation tendency from the air. In-situ and ex-situ techniques were used to grow copper (II) benzene-1,3,5-tricarboxylate (Cu-BTC MOF) MOFs on wool textiles (Abdelhameed et al., 2017). Among two different synthesised materials, i.e. in-situ- Cu-BTC MOF and ex-situ- Cu-BTC MOFs, the former one was found to have the highest adsorption capability against phenol. Furthermore, in-situ Cu-BTC MOF demonstrated a maximum removal efficacy and adsorption capability of 90% and 333 mg/g, respectively.

Ren et al. (2019) grown Zeolitic imidazolate framework-8 (ZIF-8) and Bovine serum albumin (BSA) mineralised ZIF-8 onto hydrophilic *Juncus effusus* fibres (JEFs) fibres via in-situ approach by utilising magnetic stirring and biomimetic mineralisation technique, respectively, to synthesise two different composite materials named as ZIF-8/JEFs and ZIF-8-BSA/JEFs. Among different JEFs based adsorbing materials,

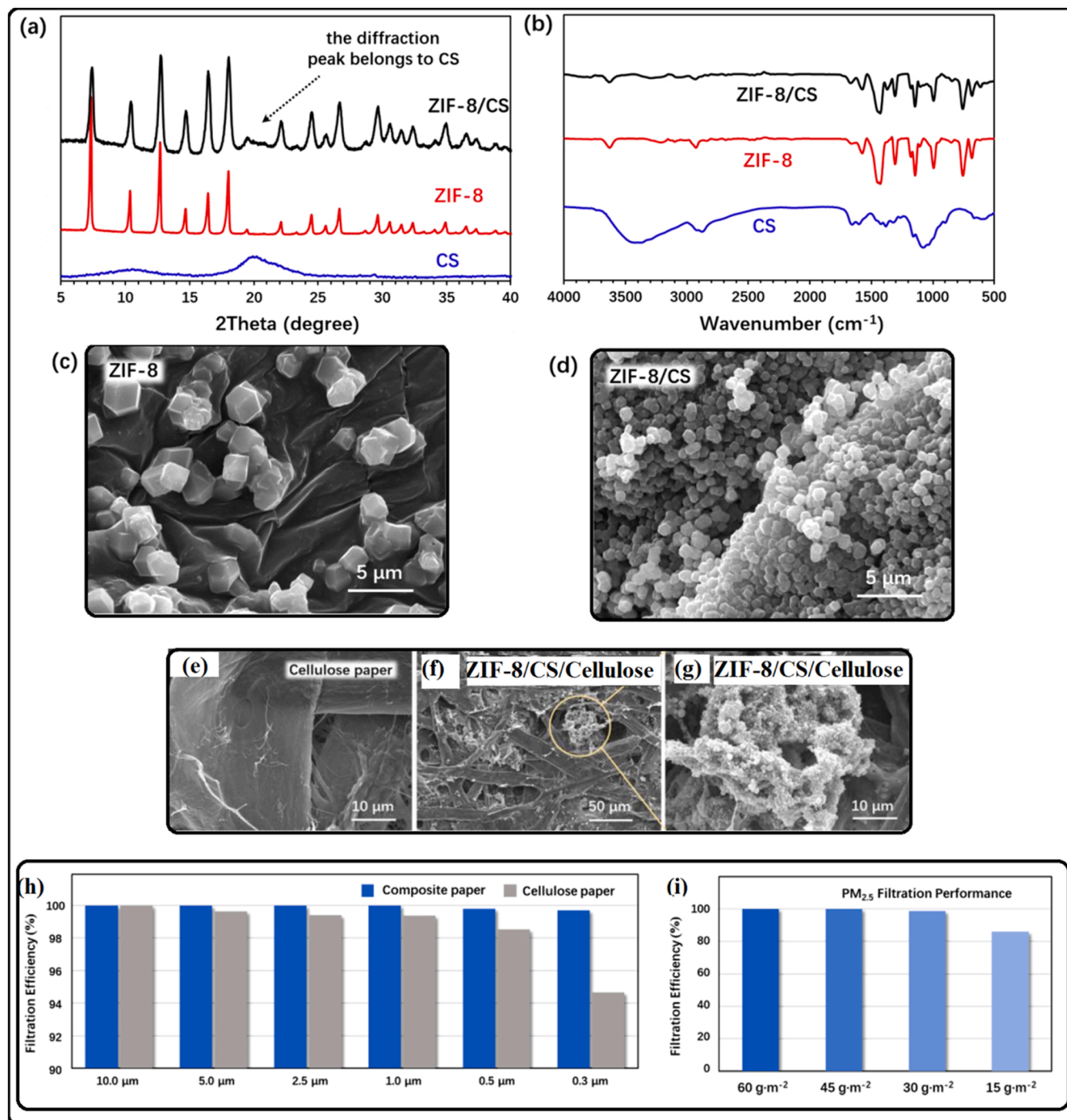


Fig. 6. (a) XRD pattern (b) FTIR spectra of CS, ZIF-8 and ZIF-8/CS (c) SEM image of ZIF-8, (d) SEM image of ZIF-8/CS, (e) SEM image of cellulose paper, (f&g) SEM images of ZIF-8/CS/Cellulose at different magnifications, (h) filtration efficacy of composite filters having a weight of 60 $\text{g}\cdot\text{m}^{-2}$ and (i) permeability of different paper materials (Nie et al., 2020). Reprinted with permission from Ref. Nie et al. (2020) Copyright 2020, Elsevier.

ZIF-8-BSA/ JEFs composites were highly efficient for removing HCHO. About 99.79% HCHO removal efficiency was noted at a higher concentration (100 $\text{mg}\cdot\text{m}^{-3}$) within 90 min, whereas at low concentration (10 $\text{mg}\cdot\text{m}^{-3}$), 100% removal efficiency was recorded within 15 minutes. Further, among different adsorbents, the maximum removal efficiency was observed for ZIF-8-BSA nanocrystals (100%) followed by ZIF-8-BSA/ JEFs composites (99.79%), ZIF-8/JEFs (94.68%) and JEFs (72.15%). From the above results, it is clear that doping of BSA nanocrystals onto JEFs has resulted in a significant enhancement in HCHO removal efficiency. Further, adsorption capacities for JEFs, ZIF-8/ JEFs,

and ZIF-8-BSA/ JEFs were found to be 36.075, 47.338, and 49.893 $\text{mg}\cdot\text{g}^{-1}$, respectively. In another study, an in-situ approach was also used to create carboxymethyl cellulose sodium (CMC)-HKUST-1 foam to remove VOCs such as toluene vapour. CMC-HKUST-1 foam (with 55 wt% stacking) have been found to have 5.2 times and 1.4–1.7 times better toluene removal efficiency than pure HKUST-1 and MIL-101 (Cr), respectively, under low pressure (P/P₀) of 0.0026 (Cui et al., 2019).

ZIF-based membranes are the most appealing among many MOFs for CO_2 adsorption (Guan et al., 2020). The preferred adsorption of carbon

Table 8Removal of VOCs and CO₂ from air using MOFs/Cellulose based materials.

Composite Material	Synthesis approach	Type of Form	% wt. loading of MOFs	Testing conditions (feed pressure/ temperature/ flow rate)	Selectivity	Uptake/ Permeability	Ref.
ZIF-L/TOCNFs	In -situ	foam	50	1 bar, 100 ml flow rate	10.5 (CO ₂ /N ₂)	0.90 mmol/g	(Valencia and Abdelhamid, 2019)
ZIF-8-NH ₂ /BC	In -situ	foam	34.8	1 bar, 25 °C	22 (CO ₂ /N ₂)	1.63 mmol/g	(Ma et al., 2021)
EC/ZIF-8/GO	Casting approach	membrane	20	0.2 MPa, 25–65 °C	33.4 (CO ₂ /N ₂)	203	(H. Yang et al., 2019; S. Yang et al., 2019; K. Yang et al., 2019)
CA/HKUST-@GO _{0.12} (means 0.12 wt% loading)	Phase Inversion technique	membrane	0.40	0.15 MPa	—	183.51 L/m ² h	(H. Yang et al., 2019; S. Yang et al., 2019; K. Yang et al., 2019)
NH ₂ -ZIF-8)/CTA	Casting approach	membrane	10	5 bar, 25 °C	33.4 (CO ₂ /CH ₄)	218	(Raza et al., 2020)
UiO-66-NH ₂ /CNFs-COOH	vacuum filtration	membrane		2 bar, 25 °C	46 (CO ₂ /N ₂)	139	(F. Zhang et al., 2018; R. Zhang et al., 2018; Y.-G. Zhang et al., 2018; X.-F. Zhang et al., 2018, p. 2)
UiO-66/CNCs	Solvothermal	Nano composites	98	1.2 bar, 0 °C		Polyethyleneimine functionalized CNCs showed better CO ₂ uptake than CNCs-COON and CNCs-SO ₃ H	(Policicchio et al., 2020)
ZIF-90/TOCNFs	In-situ cum vacuum filtration	Film coated on filter paper	44.2	25 °C, 0.2 MPa	120 (CO ₂ /CH ₄)	995 × 10 ³ mol/m ² sPa	(Matsumoto and Kitaoka, 2016)
ZIF-8/CNFs	In-situ synthesis followed by vacuum filtration	Membrane	30	25–85 °C, 0.1–0.4 MPa	36.2 (CO ₂ /CH ₄); 45.5 (CO ₂ /N ₂)	550	(Jia et al., 2020)
MOF-5/PCC filled cellulose paper	In-situ technique	Filter Paper	75	—	potential to adsorb different gaseous i. e H ₂ , CO ₂ , CH ₄ , etc. was measured	—	(Yang et al., 2017)
NH ₂ -MIL-53 (Al)/hollow CA membrane	Spinning technique	Membrane	15	3 bar, 25 °C	16 (CO ₂ /CH ₄); 12 (CO ₂ /N ₂)	11.9 GPU	(Mubashir et al., 2019, p. 53)
NH ₂ -MIL-53 (Al)/CA	Solution blending technique	Membrane	15	3 bar, 25 °C	28.7 (CO ₂ /CH ₄); 23.4 (CO ₂ /N ₂)	52.6	(Mubashir et al., 2018)
SAMH-3/CA	In situ	membrane	6	—	(CO ₂ /CH ₄)29.71	11.59 ± 0.39 Barrer	(Takahashi and Kuroda, 2011)

dioxide by MOFs is dependent on (i) the chemical/physical interaction between the MOFs' surface and carbon dioxide and (ii) the size limitation. Membranes or composites based on cellulose-MOFs have been discovered to have a higher degree of CO₂ adsorption. Valencia and Abdelhamid (Valencia and Abdelhamid, 2019) used an in-situ technique to build zeolite imidazolate frameworks (ZIF-L) onto TOCNFs. The synthesised foams (ZIF-L/TOCNFs) were found to be highly porous, have a density in between 19.18 and 37.4 kg/m³, and be extremely light. ZIF-L stacking was varied from 21 to 50 wt%, and it was found that increasing ZIF-L stacking improves the ability of the resultant composite membrane for adsorption of CO₂.

Using an ex-situ synthesis technique, varying amounts of ZIF-L nanosheets (10–40 wt%) were integrated into 2 wt% carboxymethyl cellulose (CMC) via a one-step ecologically water-based and vacuum-dried process (F. Zhang et al., 2018; R. Zhang et al., 2018; Y.-G. Zhang et al., 2018; X.-F. Zhang et al., 2018). When compared to traditional polymer matrices, CMC has numerous benefits, like uniform distribution of ZIF-L nanosheets, strong adhesion with the framework and water-based production. The capability of resulting composite sheets to separate different mixes of gaseous H₂/CO₂, H₂/N₂, CO₂/CH₄ and N₂/CH₄ was examined, and it was determined that CMC–ZIF-L with 30-wt% ZIF-L nanosheet has the best gas separation performance. Amino-functionalized ZIF-8 (ZIF-8-NH₂) was wrapped uniformly onto bacterial cellulose (BC) via in situ approaches for the development of highly porous and compatible adsorbent foam (ZIF-8-NH₂/BC) (Ma et al., 2021). Better compatibility between cellulose and MOFs has been assigned because of the chelating effect between zinc metal ions and

amino groups. The synthesized foams were found to have a high adsorption capability of 1.63 mmol/g at 25 °C, and 1 bar against CO₂.

Mixed matrix membranes (MMMs) are another type of cellulose/MOFs based membrane which were also found to have better adsorbing against CO₂. ZIF-8, EC, and graphene oxide (GO) based MMMs were developed utilising the casting approach (H. Yang et al., 2019; S. Yang et al., 2019; K. Yang et al., 2019), which comprises growing of ZIF-8 on GO surface using ultrasonic waves, followed by mixing ZIF-8 grown GO with EC for casting on a plate. The growth of ZIF-8 onto GO surface was proposed to be supported by the presence of oxygen carrying functional groups in GO (Ashour et al., 2017). The goal of incorporating GO into the membrane was to simply improve the adsorbing nature of MMMs and was observed to improve as the stacking of ZIF-8 grown GO increased. At 20 wt% stacking of GO, the resulted (EC/ZIF-8/GO) MMMs have been found to have maximum CO₂ selectivity and permeability (CO₂/N₂) of 33.4 and 203.3 Barrer, respectively. Further, water flux and hydrophilic nature of CA/GO membrane can be enhanced by incorporating HKUST-1 (CA/HKUST-@GO membrane, with 0.12 wt% GO loading), which works by preventing the stacking of GO layers (H. Yang et al., 2019; S. Yang et al., 2019; K. Yang et al., 2019). In comparison to other membranes, the CA/HKUST@GO membrane was found to have the highest water permeate flux (183.51 L/m²h).

Researchers have also attempted to improve the adsorbing behaviour of MOFs/cellulose composites by introducing amine functional groups to the surface of MOFs [118–121]. Raza et al. (2020) fabricated amine functionalized ZIF-8 (NH₂-ZIF-8)/CTA membrane via the blending cum casting technique. They reported increase in CO₂ permeability with

enhancement in feed pressure, however opposite trend was noticed in case of permselectivity. The inclusion of nanofiller, on the other hand, improved gas separation performance. A handful of researchers picked NH_2 -ZIF-8 as a filler because its pore size lies in the middle of kinetic diameter of the gases to be separated (CH_4 : 0.38 nm and CO_2 : 0.33 nm) (Ashour et al., 2017; Raza et al., 2020). Amine functionalised Universitetet I Oslo MOFs nano particles (UiO-66-NH_2) were mixed with closely packed CNFs to create an $\text{UiO-66-NH}_2/\text{CNFs-COOH}$ composite membrane via vacuum filtration technique (F. Zhang et al., 2018; R. Zhang et al., 2018; Y.-G. Zhang et al., 2018; X.-F. Zhang et al., 2018, p. 2), which was found to have a 7-times greater selectivity (CO_2/N_2) than a membrane lacking CNFs. Adding an amino group to MOFs generally increases its adsorption capability as well as its selectivity. Furthermore, introduction of different surface functionalized (carboxylic acid, sulphonic acid and polyethylenimine treated) CNCs into UiO-66 had also led to significant improvement in adsorption of CO_2 when compared to pure UiO-66 (Pollicchio et al., 2020).

Zeolitic imidazolate framework-90 (ZIF-90) was grown onto surface TOCNFs through in-situ approach. The synthesised material (ZIF-90/TOCNFs) was subsequently coated on the surface of filter paper (Matsumoto and Kitaoka, 2016). The resulted material was found to have high selectivity (120) for carbon dioxide then methane. The preference for ZIF90 (which comprises of Zn(II) tetrahedral centres and 2-imidazolecarbaldehyde (ICA) linkers) may be owing to its modest pore diameter (0.35 nm), which falls in the middle of the kinetic diameters of CH_4 and CO_2 , as well as the non-covalent adhesion between CO_2 and carboxyl group in ICA. Further, the selectivity was also shown to improve as the thickness of ZIF-90 was increased. Jia et al. (Jia et al., 2020) used an in situ technique followed by a vacuum filtering procedure to achieve homogeneous growth of ZIF-8 MOFs crystals on CNFs. The ZIF-8/CNFs membrane, which was synthesised with 70% CNFs, was found to have a permeability of 550 Barrer for carbon dioxide and optimum selectivity of 36.2 and 45.5 for CO_2/CH_4 and CO_2/N_2 , respectively.

In addition, utilizing precipitated calcium carbonate (PCC) loaded cellulose papers, efforts were undertaken to create and improve the stacking of MOF-5 ($\text{Zn}_3(\text{BDC})_2$) crystals onto cellulose paper via in-situ technique (Yang et al., 2017). The PCC-filled cellulose papers resulted in a MOF-5 loading of 337.3%, which is significantly greater than virgin cellulose paper (118.5% of MOF only). The resultant material showed great potential in adsorption of different gaseous (CH_4 , N_2 , CO_2 , H_2 , etc.).

Using a mixing process, MMMs of NH_2 -MIL-53(Al) and CA were created with varying amounts of MOFs ranging from 0 to 20 wt% for CO_2 adsorption (Mubashir et al., 2018). The addition of 15% MOFs to the CA matrix increased CO_2 permeability from 15.5 to 52.6 Barrer and CO_2 selectivity over CH_4 and N_2 from 10.7 to 28.7 (CO_2/CH_4) and 8.8–23.4 (CO_2/N_2), respectively, for the resulting MMMs. In addition, the MMMs demonstrated a CO_2 plasticization resistance of 30 bar, which is 200% higher than the virgin CA membrane (only 10 bar) (Mubashir et al., 2018). The efficacy of MMMs (NH_2 -MIL-53(Al)/CA) against CO_2 has been significantly improved through further optimization of spinning conditions (Mubashir et al., 2019). MMMs showed the highest CO_2/CH_4 ideal selectivity of 16.0 and CO_2/N_2 ideal selectivity of 12.0 at optimum spinning conditions of 5.0 cm air gap distances and 12.2 m/min take up speed.

The nanoporous layered silicate AMH-3, a 3-D porous layered silicate, contains 8 membered ring (MR) pores of size 3.4 Å, has been utilized for CO_2/CH_4 and H_2/N_2 gas separations (Jeong et al., 2003). AMH-3 was initially synthesised in year 2003, however several challenging issues have been overcome during the preparation of AMH-3/polymer composite membranes having high gas separation efficiency (Takahashi and Kuroda, 2011). Recently, exfoliated primary amine intercalated porous layer SAMH flakes were utilized for fabrication of SAMH-3/CA membrane (Kim et al., 2013). The fabricated membranes showed significant enhancement in CO_2/CH_4 gas separation ability, when loading of SAMH-3 flakes was varied from 2 to 6 wt%. A

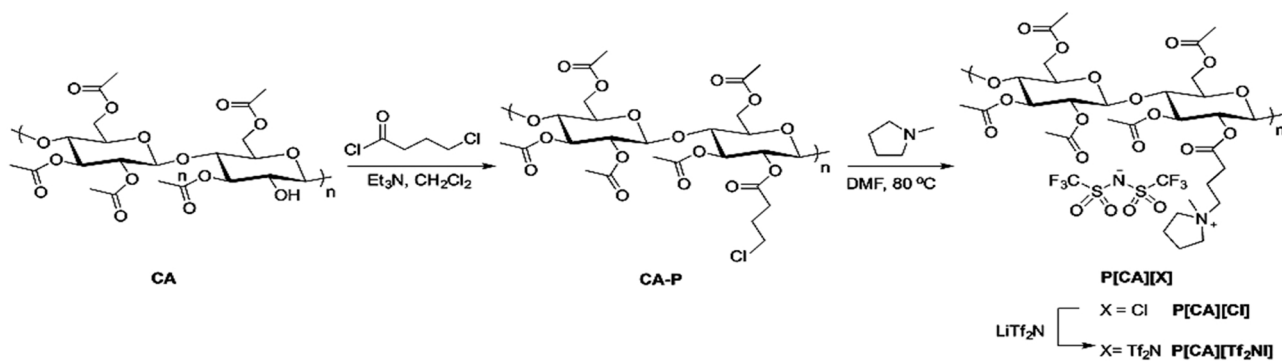
maximum permeation of 11.59 ± 0.39 Barrer was noted for CO_2 at 6 wt % loading, which is about 56% higher than permeation showed by pure cellulose acetate (7.55 ± 0.29). However, no significant improvement was observed in CO_2/CH_4 selectivity for the resulted membrane after incorporation of SAMH-3 flakes into it.

As a whole, we can conclude that adsorption of carbon dioxide or other vapours utilising a MOFs/Cellulose composite membrane has various advantages over pure MOFs or cellulose, including better adsorption abilities and permeability, as well as lower gas leakage than virgin MOFs. MOFs/Cellulose composite membranes can also be transformed into a variety of useful forms, like foams, membranes, and aerogels.

4.7. Cellulose/ionic liquids composites for air purification

Poly (ionic liquids) (PILs) materials, due to their excellent antimicrobial properties (Wang et al., 2019), flexibility, availability of numerous positively charged active chemical groups (Elshaarawy et al., 2019) and ease of processibility (Wang et al., 2019), have been highly preferred for fabrication of environmentally friendly air filters with low air resistance and highly $\text{PM}_{2.5}$ adsorbing nature (Ma et al., 2019; F. Zhu et al., 2018; M. Zhu et al., 2018; F. Zhu et al., 2018; M. Zhu et al., 2018). Zhu et al. (2020) fabricated nano fibrous filters by electrospinning of cellulose modified by PILs [prepared by a mixture of methacryloyl chloride: 0.4 gm/1-vinyl-3-butylimidazole bromide ([VBIm]Br): 1–4 gm/ 2, 2'-azobis (isobutyronitrile): 0.2 gm] using acetic acid as solvent. Improvement in removal efficiencies against $\text{PM}_{2.5}$ and PM_{10} was noted, when amount of [VBIm]Br was varied from 1 to 4 gm. With the inclusion of 4 gm of [VBIm]Br, the filtration efficacy of $\text{PM}_{2.5}$ enhanced from 27.14% to 97.94% and of PM_{10} from 94.22% to 99.65%. CTA functionalized by two different ILs namely 1-ethyl-3-methylimidazolium dicyanamide ([emim][dca]) and 1-ethyl-3-methylimidazolium tetrafluoroborate ([emim][BF₄]) demonstrated a huge potential for CO_2/N_2 permeability and CO_2/CH_4 solubility selectively because of higher affinity of doped ILs towards CO_2 in comparison to CH_4 (Lam et al., 2016). Further, CTA doped with 55 wt% [emim][BF₄] showed a solubility selectivity of 22 for CO_2/CH_4 , which was found to be much higher than values (9.1) observed at 35 °C for pure CTA. Attempt was also made for the synthesis of a new class of CA-based PIL thin film polymer membrane for isolation of CO_2 (Nikolaeva et al., 2018). CA was initially treated with 4-chlorobutylchloride in the presence of triethyl amine and dichloromethane to give an intermediate CA-P; which was subsequently reacted with *N*-methylpyrrolidinium to give the corresponding PIL (P[CA][Cl]). Using Bis(trifluoromethylsulfonyl)imide lithium salt, the chloride anion in P[CA][Cl] was ion exchanged to bis (trifluoromethylsulfonyl)imide, P[CA][Tf₂N] (Scheme 1). The CO_2 separation performances of P[CA][Tf₂N] (5.3 μm thickness) was evaluated by gravimetric gas sorption, single gas time-lag experiments, and mixed-gas separation experiments and the results were compared to the pure CA. The P[CA][Tf₂N] PIL showed an ideal CO_2/N_2 sorption selectivity up to 10 bar, increased CO_2 permeability for the CO_2/CH_4 and improved ideal CO_2 selectivity for the CO_2/N_2 during single gas transport tests and two-fold higher CO_2 flux compared to conventional CA during mixed-gas permeation tests.

Same group, in a different study, surface tailored the intermediate CA-P with different types of IL pendants like 1-methylimidazol (Im), 1-methylpyrrolidine (Pyr), 2-hydroxyethyl dimethylamine (HEDMA) and a mixture of the relative wt% (75:25; 50:50; & 25:75) of Pyr and HEDMA (Nikolaeva et al., 2021). Further, in the synthesised materials i.e P[CA][Cl], P[CA-Pyr][Cl], P[CA-HEDMA][Cl], P[CA-75Pyr25HEDMA][Cl], P[CA-50Pyr50HEDMA][Cl] and P[CA-25Pyr75HEDMA][Cl], the chlorine ion was subsequently ion exchanged with lithium salt (LiTf₂N) to give P[CA][Tf₂N], P[CA-Pyr][Tf₂N], P[CA-HEDMA][Tf₂N], P[CA-75Pyr25HEDMA][Tf₂N], P[CA-50Pyr50HEDMA][Tf₂N] and P[CA-25Pyr75HEDMA][Tf₂N]. These cell/poly(IL) were then solvent casted into thick membrane of thickness 4–6 μm. After that, the cell/poly(IL) mixture was



Scheme 1. Conversion of CA into P[CA][Tf₂N] through IL sites incorporation and subsequent anion metathesis. Scheme copied from Ref. Nikolaeva et al. (2018) Copyright 2018, Elsevier.

solvent cast into a thick membrane with a thickness of 4–6 μm . The CO_2/N_2 selectivity of all monofunctional cells/poly(IL) initially rises with increasing CO_2 partial pressure in the feed mixture up to 50 vol% CO_2 . However, as CO_2 concentrations further increased, there was a modest decrease in CO_2/N_2 separation. Furthermore in case of bi-functional cell/poly(IL), with increasing HEDMA content, the resulting membrane demonstrated a decrease in CO_2 and N_2 permeability, as well as an initial fall and then steady increase in CO_2/N_2 selectivity. Thus HEDMA content linked to the CA backbone controls the overall separation process of bifunctional cationic pendants based

cell/poly(IL).

Further, four types of solid ionic liquids (SoIL) with and without cellulose support (cellulose contents varied from 70 to 80 wt% i.e. from 30: 70 wt. ratio of SoIL/cellulose to 20:80 wt. ratio), namely tetraethyl ammonium acetate ([N₄₄₄₄][Ac], methyltriethyl ammonium acetate ([N₁₈₈₈][Ac], tetra-octylammonium bromide ([N₈₈₈₈][Br] and 1-butyl-4-methylimidazolium bromide ([Bmim][Br] were utilized for adsorption of CO_2 (García-Gutiérrez et al., 2019). Adsorbents with a 25:75 wt. ratio of SoIL/cellulose demonstrated the best adsorbing performance among varied wt% SoIL supported cellulose. In addition, adsorbing trend

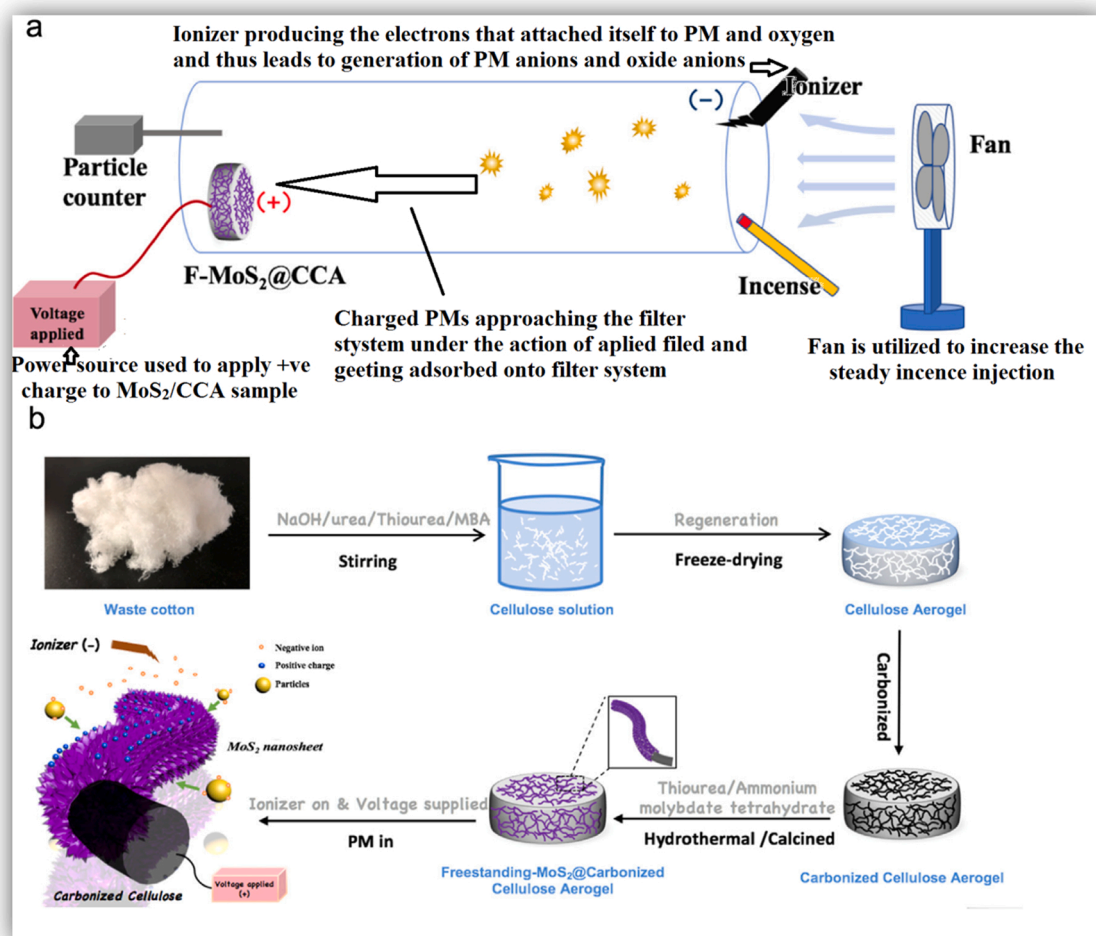


Fig. 7. Schematic depiction of (a) home-built PM adsorption set up and (b) the in-situ synthesis of F-MoS₂/CCA and PM adsorption process (Xie et al., 2021). Reprinted with permission from Ref. Xie et al. (2021) Copyright 2021, Elsevier.

observed among four types of SoIL supported cellulosic samples was noticed to be as: Cell/[N₄₄₄][Ac] (31.6 g CO₂/kg adsorbent) > Cell/[N₁₈₈][Ac] (30.6 g CO₂/kg adsorbent) > Cell/[Bmim]Br (28.8 g CO₂/kg) > Cell/[N₈₈₈][Br] (19.2 g CO₂/kg adsorbent). In comparison to cellulose samples without SoIL support, a significant increase in adsorption performance was observed for SoIL supported samples.

4.8. Carbonized cellulose-based filter system

4.8.1. Particulate matter removal

Due to traditional filtration fabrics' common drawbacks, such as their larger size, high energy consumption, and large ΔP , researchers are now concentrating their efforts on aerogel-based filtration systems (Mao et al., 2019; F. Zhang et al., 2018; R. Zhang et al., 2018; Y.-G. Zhang et al., 2018; X.-F. Zhang et al., 2018). To develop a carbonised cellulose aerogels-based filter system, cellulose aerogels were initially synthesised from cotton cellulose using a freeze-drying technique, then carbonised at 800 °C for 200 min to produce carbonaceous cotton aerogels (CCA) (Fig. 7b). Finally, the CCA was hydrothermally anchored with MoS₂ to create a better absorbance MoS₂/CCA filtering system (Xie et al., 2021). The primary goal of carbonization was to recycle cotton materials that would otherwise be thrown away. In a homemade PM adsorption test system, the developed MoS₂/CCA was evaluated for its ability to adsorb the PMs (Fig. 7a). The workings of the developed system are depicted in Fig. 7. The produced composite air filter had a removal effectiveness of over 99.95% and 99.91% for PM₁₀ and PM_{2.5}, respectively, and was shown to have a strong adsorbing ability for up to 600 min (PM₁₀ > 99.1%; PM_{2.5} > 98.6%). The composite air filter also demonstrated outstanding re-usability, with an adsorption effectiveness of > 97.80% over five successive cycles.

4.8.2. Gas adsorption

Nitrogen-doped cellulose-based porous carbon materials were synthesized by hydrothermal carbonization of microcrystalline cellulose with melamine mixture of different ratios at a certain temperature for 5 h followed by heating of hydrothermal products with KOH (1:3) at 800 °C, under nitrogen atmosphere for 1 hr (Tan et al., 2021) (Fig. 8). Microporous of maximum specific surface area of 1703 m²/g, and volume of 0.65 cm³/g with nitrogen-doping constitution of 1.68 atom % were obtained, when ration of melamine and microcrystalline cellulose

was kept at 1:1 and hydrothermal carbonized temperature was 270 °C. At 25 °C, the carbonised products had CO₂ adsorption efficiencies of 7.34, 5.42, 3.79, and 4.56 mmol/g, respectively, under pressures of 0.4, 0.3, 0.2, and 0.1 MPa, respectively. Zhuo et al. (2016) made two types of carbon aerogels, CO₂ activated carbonized cellulose aerogel (Cell-CO₂) and nitrogen activated carbonized cellulose aerogel (Cell-N₂), by heating cellulose aerogels to 800 °C at a rate of 3 °C/min under CO₂ and N₂ atmospheres, respectively, and then holding at 800 °C for 2 hr. Their results indicated higher specific area (1364 m²/g), micro volume (0.37 cm³/g) and total volume (1.42 cm³/g) for Cell-CO₂ than Cell-N₂ (specific area: 859 m²/g; micro volume: 0.32 cm³/g and total volume: 0.75 cm³/g). We can deduct from the foregoing findings that the CO₂ activation process produces more meso- and micro-pores, thus resulting in a higher specific surface area than N₂ activation process. Further, Cell-CO₂ displayed a higher adsorption capacity (3.42 mmol/g) for CO₂ than Cell-N₂ (3.01 mmol/g), which may be related to its higher specific area. Solvothermal conversion of cotton into highly porous carbonaceous materials (SCM) was also carried out using varying concentration (0.04–7.0 wt%) of p-toluene sulfonic acid (PTSA) catalyst (Sivadas et al., 2015). The optimum CO₂ removal efficiency (2.28 mmol/g at 1 bar pressure and 0 °C) was noted when natural cotton was treated with 0.04 wt% PTSA. From, above discussion we can conclude that carbonaceous materials have tremendous potential to remove CO₂ or to adsorb the PMs and are far superior to other cellulosic materials.

4.9. Cellulose/polymer matrix-based polymer composites for air purification

Dual-size TEMPO oxidised CNFs/PVP nanofibers composites of submicron and nano size were spun onto HEPA filter via one-step electrospinning technique to obtain a multilayer air filter (Balgis et al., 2017) (Table 9). The wt. percentage of CNFs and PVP was varied from 0% to 0.2% and 0–10% respectively, to obtain the nano fibers composite of optimum properties. Nodoubt, the optimized multilayer air filter Cell/PVP (0.2/8 wt%) showed a filtration efficiency (86.4%) lower than that of pure HEPA (99.99%). However, its quality factor value (0.117 Pa⁻¹), was found to be 10 folds higher than that of commercial HEPA (0.046–0.061 Pa⁻¹).

CNCs/PVA electrospun nano filter were synthesised with different percent stacking of CNCs ranging from 5 to 20 wt% (Q. Zhang et al.,

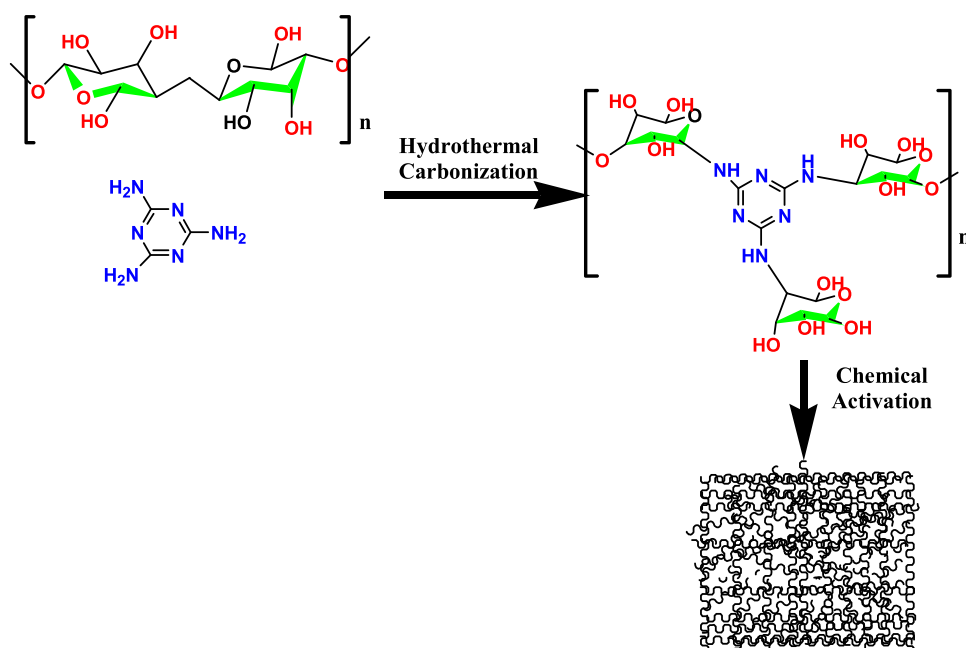


Fig. 8. Nitrogen-doped cellulose-based porous carbon reaction process (Tan et al., 2021).

Table 9

A comparative data of efficiency of cellulose/ matrix-based filters in removal of PM.

Name	Type of filter	Amount of cellulose material added	Synthesis technique	Removal Efficiency	Flow Rate (cm/s)	ΔP (Pa)	Ref
Dual size CNFs/PVP nano fiber composites	Multilayer filter	0.2 wt%	Electrospinning	PM _{0.3-0.5} : 86.4%	5.3	17	(Balgis et al., 2017)
CNCs/PVA	Filter	20 (80 mg m ⁻² wt.)	Electrospinning	PM _{2.5} : 99.1%	0.2 m/s	91	(Q.Q. Zhang et al., 2019; H. Zhang et al., 2019)
Heated CNCs/PVA	FILTER	20	Electrospinning	PM _{2.5} : 96%	0.2 m/s	60	(Zhang et al., 2020)

2019; H. Zhang et al., 2019) (Table 9). An optimum removal efficiency of 99.1% was achieved for PM_{2.5}, when electrospinning time was enhanced for 8 min (means a thicker filter) and CNCs stacking was maintained at 20 wt%. A decrease in zeta potential value with the enhancement in percent stacking of CNCs was also noted. Further, same group extended their efforts to synthesise reusable CNCs/polyvinyl alcohol (PVA) filter with remarkable PMs removal performance via electrospinning followed by facile heat treatment at 140 °C for 5 min (Zhang et al., 2020). The main goal of heat treatment was to turn water soluble CNCs/PVA air filters into entirely water proof or washable air filters while avoiding the use of any cross-linking agent. The prepared air filter maintained a removal efficiency > 95% for PM_{2.5}, even after being recycled 5 times, showing that regeneration had no effect on the removal efficiency. Furthermore, the ΔP was noticed to enhance by 156.7% after 5 cycles, yet it was still < 100 Pa. Also, the Qf value of recycled filter remained > 0.035 Pa⁻¹ after 5 cycles, which is comparable superior to other commercial filters.

4.10. Cellulose/polyoxometalates based air filter system

Polyoxometalates (POMs) are metal-oxygen cluster anions. These anions could be of type $M_xO_y^{m-}$ or $X_2X'_zM_xO_y^{n-}$, where M stands for W, Mo, or V, X for B, P, Ge, or Si, and X' for a transition metal (Pope and Müller, 2002; H. Yang et al., 2019; S. Yang et al., 2019; K. Yang et al., 2019). POMs have been recognised as pretty desirable redox catalysts in heterogeneous and homogeneous media because of their higher hydrolytic and thermal stability, configurable redox characteristics and better solubility in different solvents (Mizuno and Misono, 1998; Vazylyev et al., 2005).

POMs can be grown on a variety of inorganic and organic supports for gas or liquid phase heterogeneous catalytic purposes without contaminating the downstream process.

Various transition metal substituted POMs such $[SiW_{11}CeO_{39}]^{4-}$, $[PW_{11}CoO_{39}]^{5-}$ attached to silica and polyvanadotungstates ($PW_9V_3O_{40}$, $PW_8V_4O_{40}$, and $PW_6V_6O_{40}$) (Guo et al., 2014; Kholdeeva, 2004; Kholdeeva et al., 2005) have been successfully employed for oxidising formaldehyde to formic acid under mild conditions (20–40 °C).

Cellulose/silica hybrids (CSH) coated with Keggin-type vanadium heteropolyanions (POM-CSH) have been prepared via sol gel technique and subsequently utilised for oxidation of VOCs (C₅–C₁₁) present in polluted air (Gamelas et al., 2012). The addition of silica was intended to protect cellulose from oxidative ageing while also avoiding direct contact with POMs. While biodegradable cellulosic pulp served as catalyst support and allowed the production of less dense, highly porous and reusable material that could be used as fluffy pulp or on a paper surface.

The oxidising capability of $[PV_xMo_{10}O_{40}]^{(3+x)-}$ ($x = 1, 2$) was found to be comparable to that of UVPCO systems. During oxidation process the species V(V) gets reduced to V(IV) by interaction with VOCs and then again re-oxidised to V(V) by air molecular oxygen (Gamelas et al., 2012). Anticipated from known redox characteristics of POMs, the re-oxidation of $[PV_2Mo_{10}O_{40}]^{5-}$ was found to be quicker than $[PVMo_{11}O_{40}]^{4-}$. Furthermore, the aerobic oxidation of formaldehyde was carried out by utilising cellulose/aminopropyl coated silica hybrid materials functionalized with α -isomer of $[SiW_9O_{37}RuIII_4(H_2O)_3Cl_3]^{7-}$ (Gamelas et al., 2016). The highest formaldehyde

adsorption per kg of hybrid material coated with Ru-POMs (approx. 1.4% w/w) was noted to be 1.10 g/min. The ruthenium POMs found to be a reversible redox catalyst, retaining its catalytic activity even after more than 400 turnovers.

4.11. Cellulose-based photonic films for detection of pollutants

Photonic stimuli-responsive functional materials have sparked a lot of attention among other types of stimuli-responsive materials because of their structural colours respond to multiple sources of stimulus and can be distinguished with the naked eye (Huang et al., 2020; Kolle and Lee, 2018; Lu et al., 2016). CNCs/CNFs have played an increasingly essential role in the development of stimuli-responsive functional materials because of the availability of modifiable functional groups (OH), high specific area, low density, plentiful able pore structure, and biocompatible/biodegradable nature. The existence of modifiable functional groups on cellulose's surface makes it simple to enrich it with stimulus response sources. Above a certain concentration, negatively charged CNCs showed phase separation and create an equilibrium ordered chiral nematic phase (Araki et al., 2000; Gray, 2016). Furthermore, the chiral nematic order may be preserved after drying by evaporation-induced self-assembly (EISA), and exhibit structural colors because of their maximal reflection of light. Zhao et al. (2020) created a unique chiral nematic CNCs film with a dual sensitivity to formaldehyde and humidity. It was discovered that this film may respond in a reversible manner to formaldehyde gas and environmental humidity variations. This type of material functions as a sensor and can be used to detect formaldehyde in a humid atmosphere since it changes colour from invisible to visible under these conditions (Fig. 9). The dual response of film is attributed to the synergistic effect of collaboration and rivalry between formaldehyde and water molecules by establishing physical cross-linking networks through hydrogen bonds among formaldehyde, water and CNC.

4.12. Cellulose/metal oxide-based filter system

4.12.1. Role of cellulose-based photo-catalyst composite for degradation of VOCs and CO₂

Photo-catalytic degradation of VOCs or CO₂ is a cost-effective, safe, and environmentally friendly method of air filtration. Its numerous advantages, including ease of use, low power consumption, lack of secondary emissions, and high efficacy, make it perfect for long-term use in modern civilization (Guan et al., 2021; Ikram et al., 2021; Sudha and Sivakumar, 2015). Most extensively used photo-catalysts include WO₃ (Eg: 2.8 eV) (Widiyandari et al., 2016), TiO₂ (3.2 eV) (Dette et al., 2014), ZnO (3.37 eV) (Oliva, 2001), and CdS (2.42 eV) (Kamarulzaman et al., 2015), although these materials only have photo-catalytic activity when exposed to UV light or are partially active in visible light.

To avoid secondary contamination during the use of nano photo-catalysts for VOC degradation, they are usually doped onto cellulosic materials (Dong et al., 2018; Hua et al., 2020; Zou et al., 2019). Numerous techniques are utilized to produce nano photo-catalyst onto different fabric substrates, which include magnetron sputtering (Miao et al., 2015), sol-gel-impregnation (Uddin et al., 2008), hydrothermal treatment (Meng et al., 2014), suspension polymerization (Jiang et al.,

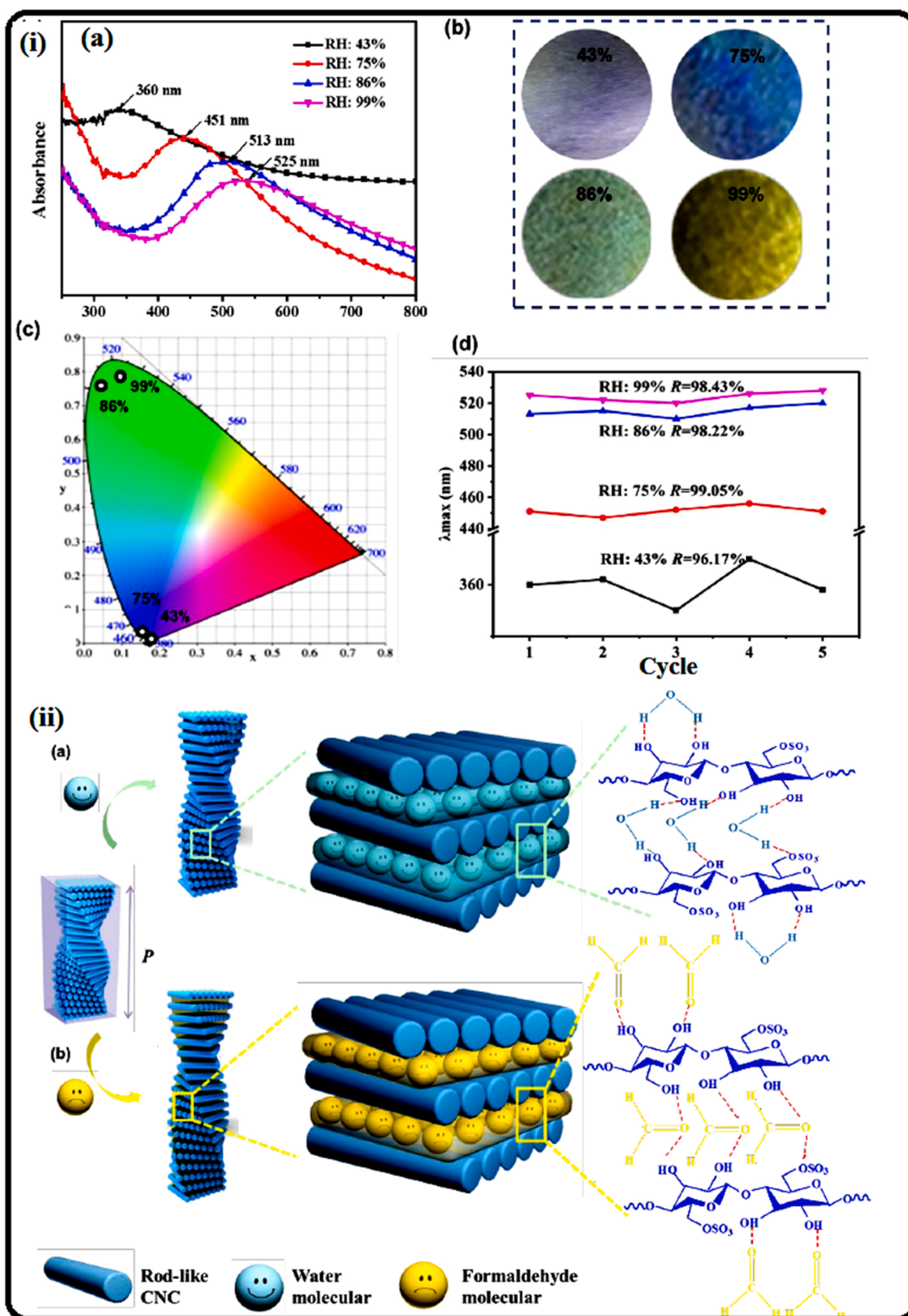


Fig. 9. Dual response of photonic crystals (Zhao et al., 2020). Reprinted with permission from Ref. Zhao et al. (2020) Copyright 2020, ACS.

2011), hydrolysis (Dong et al., 2018), plasma treatment (Yuranova et al., 2007), cross-linking methods (Xu et al., 2015) and emulsion electrospinning (Kwon et al., 2021). Among aforementioned techniques the hydrothermal approach is most permitted as it provides the perfect control of nano photo-catalyst particle size and phase composition. Ag was doped onto Cellulose/ZnO, synthesised via ultrasonic cum hydrothermal technique, to improve the photo-catalytic activity of ZnO, which was then used to degrade benzene and their derivatives and phenol on exposing with UV radiations (Zou et al., 2019). When exposed with 6W of UV radiation, the synthesised materials (Cellulose/ZnO/Ag) offered a degradation efficiency of 19.5%, 9.7%, 15.8%, 84%, and 100% against benzene, methyl benzene, ethyl benzene, biphenyl, and phenol, respectively. The cellulose acts as a VOC adsorbent while simultaneously acting as a support for the photo-catalyst. A facile hydrothermal technique was also utilised for in-situ coating of TiO₂ onto CFs. The synthesised TiO₂/CFs nano composites photo-catalytically degraded CO₂ to methane and carbon monoxide with maximal yields of 50.8 ppm/g and 194.0 ppm/g, respectively, when exposed to UV radiation for 6 hr. Furthermore, composition of rutile, brookite and anatase phases of TiO₂ was finely tuned by adjusting concentration of HCl acid.

Ethyl acetate (EA) doped TiO₂ photo-nanocatalyst (EA/TiO₂) was synthesised utilising electrospinning technique (Kwon et al., 2021). The fabricated nanofibers web showed a removal efficiency of 99.5% for PM_{0.5} and photo-catalytic degradation of 78.6% against NO. Furthermore, following doping TiO₂, which was mostly discovered to include anatase phase, a significant increase in hydrophobic character (CA-140 °C) was observed. BC acetate doped with TiO₂ nano composites after calcinations at different temperatures 500, 600, 700 and 900 °C, were tested for their photo-catalytic ability to degrade phenol under UV illumination (Lu et al., 2013). The maximum photo-degradation abilities were found for the samples calcinated at 700 °C (96.4%) succeeded by samples calcinated at 600 °C (92.3%), 500 °C (90.1%), 800 °C (81.3%)

and 900 °C (72.6%).

TiO₂/aminated CNCs photo-nanocatalyst, with high photo-catalytic activity was developed by Maimaiti et al. (2019) (Fig. 10). Their procedure involves the thionylation followed by ethylenediamine (EDA) based surface functionalization of CNCs to obtain CNCs-EDA, which was subsequently coated onto TiO₂ by employing hydrothermal technique. The prepared TiO₂/CNCs-EDA photo-catalyst was utilised to convert CO₂ to HCOOCH₃, yielding approx. 372.85 mol/g cat, about 4.5 times greater than the yield obtained with TiO₂ photo-catalyst. Same group further extended their efforts to develop a highly efficient Cu₂O/CNCs-EDA photo-nanocatalyst by wrapping cuprous oxide nano particles with CNCs-EDA (Yisilamu et al., 2018). The photo-nanocatalyst was utilised for photo-catalytic conversion of CO₂ into methanol under visible light irradiation. Furthermore, when Cu₂O/NCC-EDA (Cu₂O % = 86.56%, N % = 2.04%) photo-catalyst was used for CO₂ reduction, methanol production exceeds 1528.5 mmol/g cat, about four times greater than the production obtained with pure Cu₂O.

A visible light-responsive cellulose based photo-nanocatalyst, La_{0.3}Sr_{0.7}TiO₃/regenerated cellulose nanocomposites was developed via sol-gel method and subsequently utilized for degradation of formaldehyde (Hua et al., 2020). The maximum degradation of formaldehyde (60%) was observed at 60:40 wt. ratios stacking of La_{0.3}Sr_{0.7}TiO₃/regenerated cellulose. Further, 30% stacking of La_{0.3}Sr_{0.7}TiO₃/regenerated cellulose filler in paper showed approx. 90% formaldehyde degradation and good mechanical properties.

4.12.2. Cellulose/transition metal oxide filters for Carbon dioxide/VOCs removal

Efforts were also made for the utilisation of transition metal oxide for adsorption purpose rather than for photo-catalytic degradation of carbon dioxide (Campbell et al., 2019). Recently, transition metal oxides such as TiO₂ and Fe₃O₄ have been coated onto cellulose, extracted from

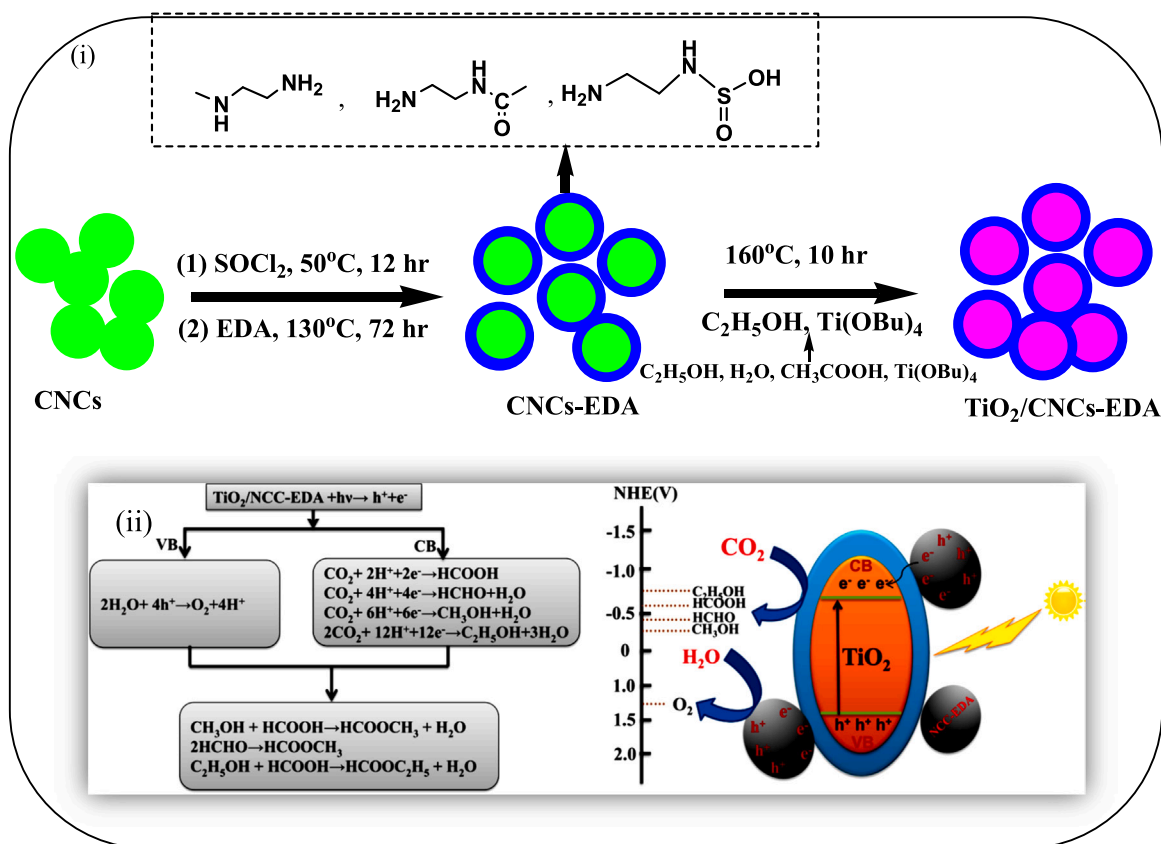


Fig. 10. (i) Scheme showing the preparation of TiO₂/CNCs-EDA photonanocatalyst and (ii) mechanism for the photocatalytic reduction of CO₂ into HCOOCH₃ (Maimaiti et al., 2019). Reprinted with permission from Ref. Maimaiti et al. (2019) Copyright 2019, Elsevier.

rice husk, utilising co-precipitation and hydrothermal techniques, respectively and subsequently utilised for separation of CO_2/N_2 and CO_2 adsorption (Campbell et al., 2019). It has been proposed that due to basic nature of TiO_2 and Fe_3O_4 , these basic groups forms covalent bonds with the acidic CO_2 through Lewis acidic-Basic interaction way and thus improve the cellulose's adsorptive properties (Fig. 11) (Campbell et al., 2019; Su et al., 2016). The synthesised Cell/ TiO_2 and Cell/ Fe_3O_4 nano composites offered a carbon oxide uptake of 184.1 mg/g and 130.6 mg/g at 30 bar and 298.15 K, respectively, which is very much higher than CO_2 (81.6 mg/g) uptake by unmodified cellulose (evaluated under the same conditions). Further, the best CO_2/N_2 selectivity result was obtained for Cell- TiO_2 (3.56 ± 0.16) followed by Cell- Fe_3O_4 (1.77 ± 0.16) and pure cellulose (0.96 ± 0.15).

Magnetite (metastable iron oxide), commonly known as black iron oxide, is biocompatible and less hazardous to humans. It has the finest magnetic properties of all transition metal oxides (Katepetch and Rujiravanit, 2011; Kim et al., 2006). Using an in-situ technique, magnetite functionalized bacterial cellulose ($\text{Fe}_3\text{O}_4/\text{BC}$) was synthesised and then utilized for the removal of polar and non-polar VOCs (Ion et al., 2015). The $\text{Fe}_3\text{O}_4/\text{BC}$ offered a better adsorbing capacity and mean adsorption rate towards isopropanol (0.730–0.805 g/g; 0.337–0.372 g/gh, respectively) and n-hexane (0.642–0.755 g/g; 0.329–0.487 g/gh) at 40 °C, when compared to pure BC (0.527–0.700 g/g; 0.198–0.263 g/gh in case of isopropanol and 0.677–0.795 g/g; 0.232–0.318 g/gh in case of hexane).

4.13. Triboelectric based removal of particulate materials

In 2012, based upon the Maxwell's displacement current laws, Wang et al. (Fan et al., 2012) first time proposed a triboelectric nanogenerator (TENG) to convert mechanical energy from the surrounding environment into electrical energy. He et al. (2018) later fabricated a fiber based TENG mask was for removal of $\text{PM}_{2.5}$. However, it can only be utilised to purify the air entering into the lungs and does not play any role in purification surrounding air. Recently, Mo et al. (2020) developed environmentally friendly self-powered PM removal system consisting of radial piston TENG, cellulose fiber air filter (CFAF) and a one-way valve for removal of PM from outdoor air (Fig. 12). The RP-TENG components of self-powered PM removal system, in addition to generation of electrical energy, its piston movement pumps contaminated air into the CFAF, where it is purified. The CFAF is made up by cellulose filter paper, while the negative and positive friction materials of the RP-TENG are composed of perfluoroethylene propylene (FEP) films and CNFs, respectively. The removal efficiency of CFAF was found to be enhanced by 1.21 and 1.27 times upon charging, showing a maximum of 86.82% and 83.78% removal efficiency for PM_{10} and $\text{PM}_{2.5}$, respectively. One of major advantage of this self-powered system is that it can be installed on any rotating vehicle, where wasted rotating mechanical energy, could be utilized to purify the outdoor environment.

Cho et al. (2021) made another breakthrough in this field in 2021, when they developed a triboelectric mechanical device based on integrated mechanical filtration owing to structural properties of

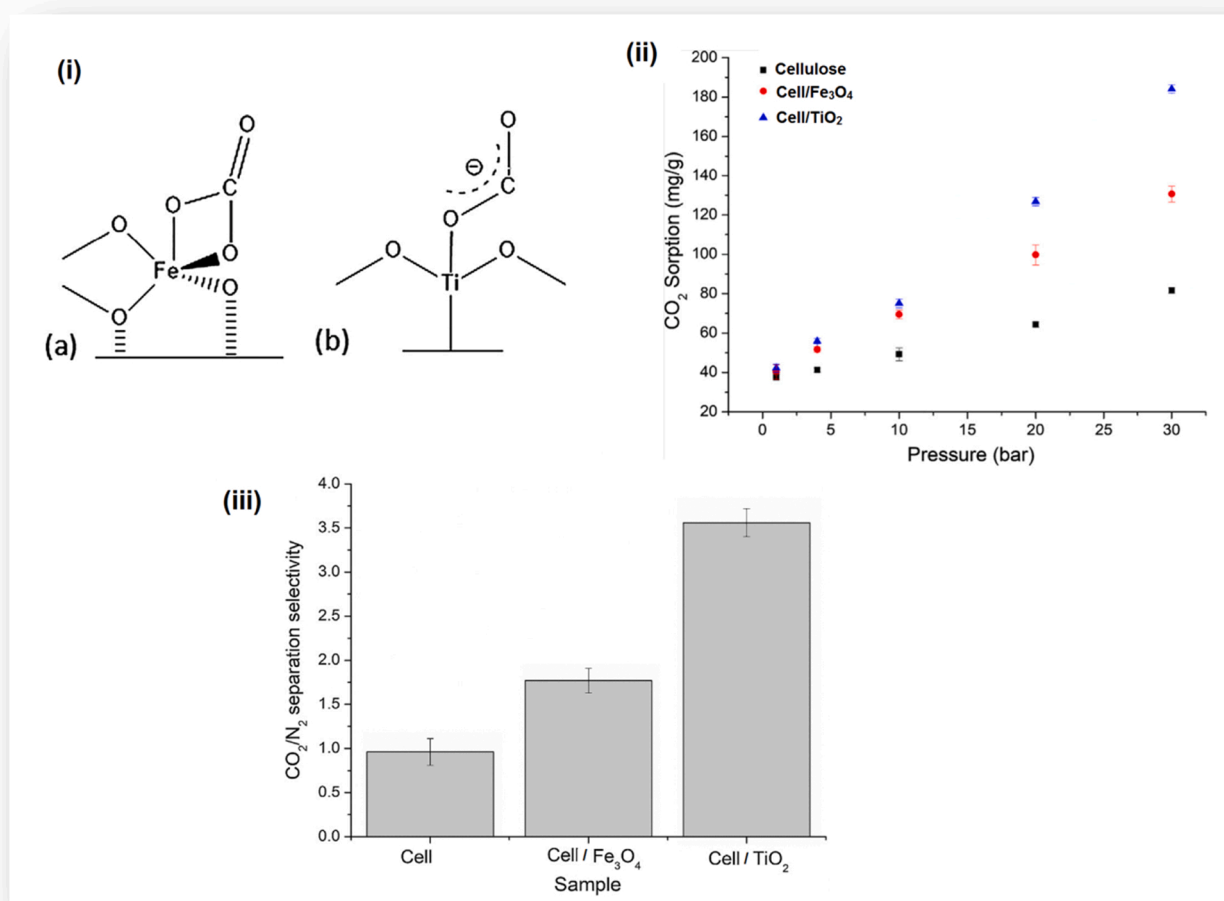


Fig. 11. (i) Scheme for adsorption between carbon dioxide and metal oxide, (ii) CO_2 sorption and (iii) separation selectivity of Cell/ TiO_2 , Cell/ Fe_3O_4 and pure cellulose (Campbell et al., 2019). Reprinted with permission from Ref. Campbell et al. (2019) Copyright 2019, Elsevier.

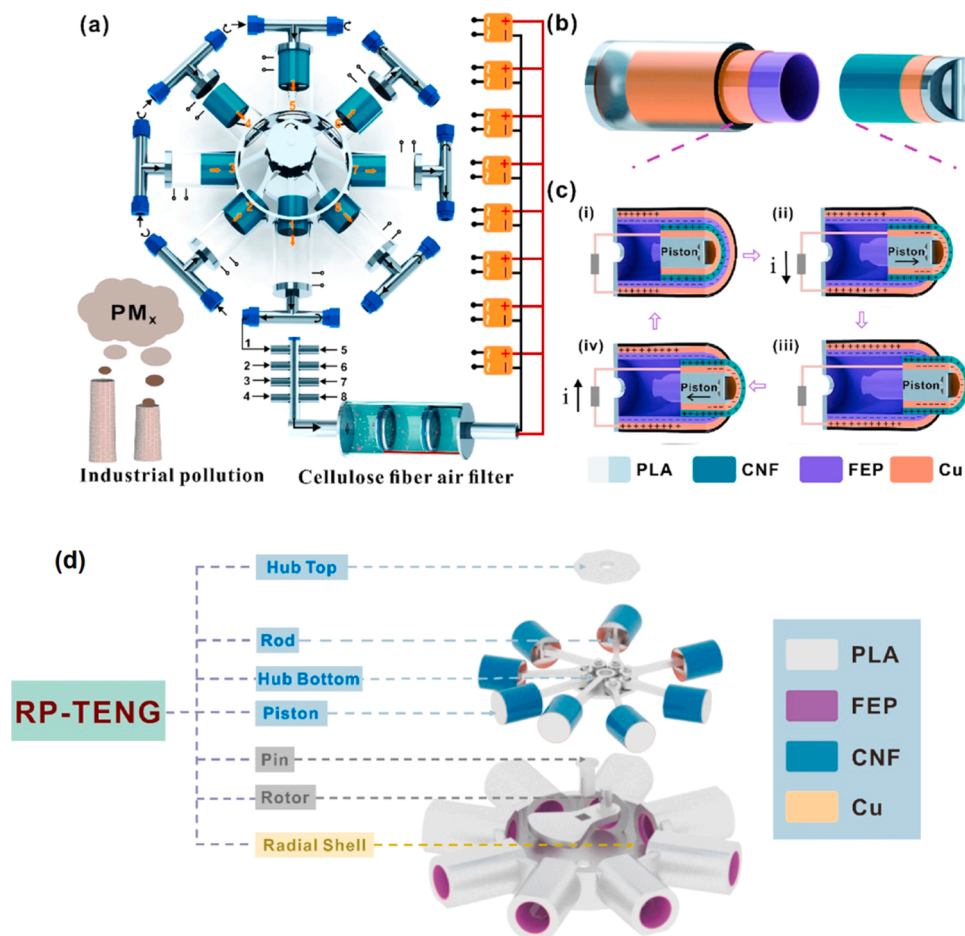


Fig. 12. Schematic diagram and working principle. (a) PM_x adsorption system; (b) One unit of TENG; (c) Working principle of RP-TENG and (d) (a) Assembly diagram of RP-TENG (Mo et al., 2020). Reproduced with permission from Ref. Mo et al. (2020) Copyright 2020, Elsevier.

electrospun ethylcellulose (EC) and electrostatic filtration stimulated by direct triboelectrification among both polytetrafluoroethylene (PTFE) spheres and EC. The detailed for triboelectric mechanical device formation, its functioning and changes in morphology and diameter of EC fibers after adsorption/or electrostatic interaction with PMs has been presented in Fig. 13. The triboelectrification of PTFE and EC resulted in the complete elimination (100%) of PM₁, PM_{2.5} and PM₁₀. As a result, this technology may be able to overcome the disadvantages of mechanical filtration while avoiding the creation of harmful chemicals.

5. Challenges and future prospectus

Several techniques are available in literature to synthesise the nano cellulose (CNCs and CNFs) from cellulose. However, the cost for production of one gm of nanocellulose will be same as for the CNTs (Cheng et al., 2020). Large-scale production of nanocellulose under anhydrous conditions is still a biggest challenge to scientists (Y. Zhou et al., 2019).

Most of the chemical techniques involve the utility of strong acids, e. g., nitric acid, sulphuric acid, hydrochloric acid (Ferreira and Rezende, 2018; Hirai et al., 2009). Nodoubt, the green approaches have also been developed for production of nanocellulose. But one should select the technique that will have little impact onto internal structure of CNCs or CNFs and will also limit the cost of extraction process (Tang et al., 2020), so that a large scale production of nanocellulose can be accomplished.

Cellulose-based materials or its derivatives can be used directly as a substrate for the in-situ or ex situ growth of MOFs or for the development of other composite filter/membranes. Direct synthesis of MOFs into pulp slurry or wood (Küsgens et al., 2009) was reported (Tu et al., 2020),

which involve only a simple pre-treatment of wood with NaOH (Tu et al., 2020). The authors have found similar type of MOFs in case of NaOH and carboxymethylated cellulosic fibers. This technique presents a green approach for the fabrication of hierarchical porous MOF/wood composites with micro- or meso-porous structures.

Increase in specific area was obtained after deposition of MOFs and same concept was used by other scientists for in-situ growth of various types of MOFs (ZIF-8, UiO-66, MOF-808, UiO-77.) onto cellulose (Bunge et al., 2018; Richardson et al., 2019). To increase the surface specific area of cellulose based filters various kind of techniques such as blending with other green materials (SPC), surface functionalization, conversion into carbonaceous materials, treatment with ILs and others have been also employed (Belaustegui et al., 2020; Lam et al., 2016; Venturi et al., 2018). Nano particles incorporation have shown low loading of MOFs onto cellulose based materials (Guo et al., 2019). Synthesis methods employed for MOFs growth and type of cellulose forms employed as substrate directly impact the loading of MOFs and thus overall performance of composite materials. So, utmost care is required for selecting appropriate cellulosic forms and synthesis techniques keeping in view application part. For example four different type of loading methods have been employed for the growth of Cu-BTC onto CMC namely layer-by-layer-water organic, layer-by-layer-organic, one-pot and two-step synthesis (Li et al., 2020). The maximum deposited amount 39.38% was noted in case of layer-by-layer-water organic followed by layer-by-layer-organic (31.49%), two step (4.23%) and one step synthesis (1.31%). Also, highest specific surface area 159 m²/g was noted for Cu-BTC@CMP synthesised by using the layer-by-layer-water organic procedure. Thus, choice of deposition technique controls the number of

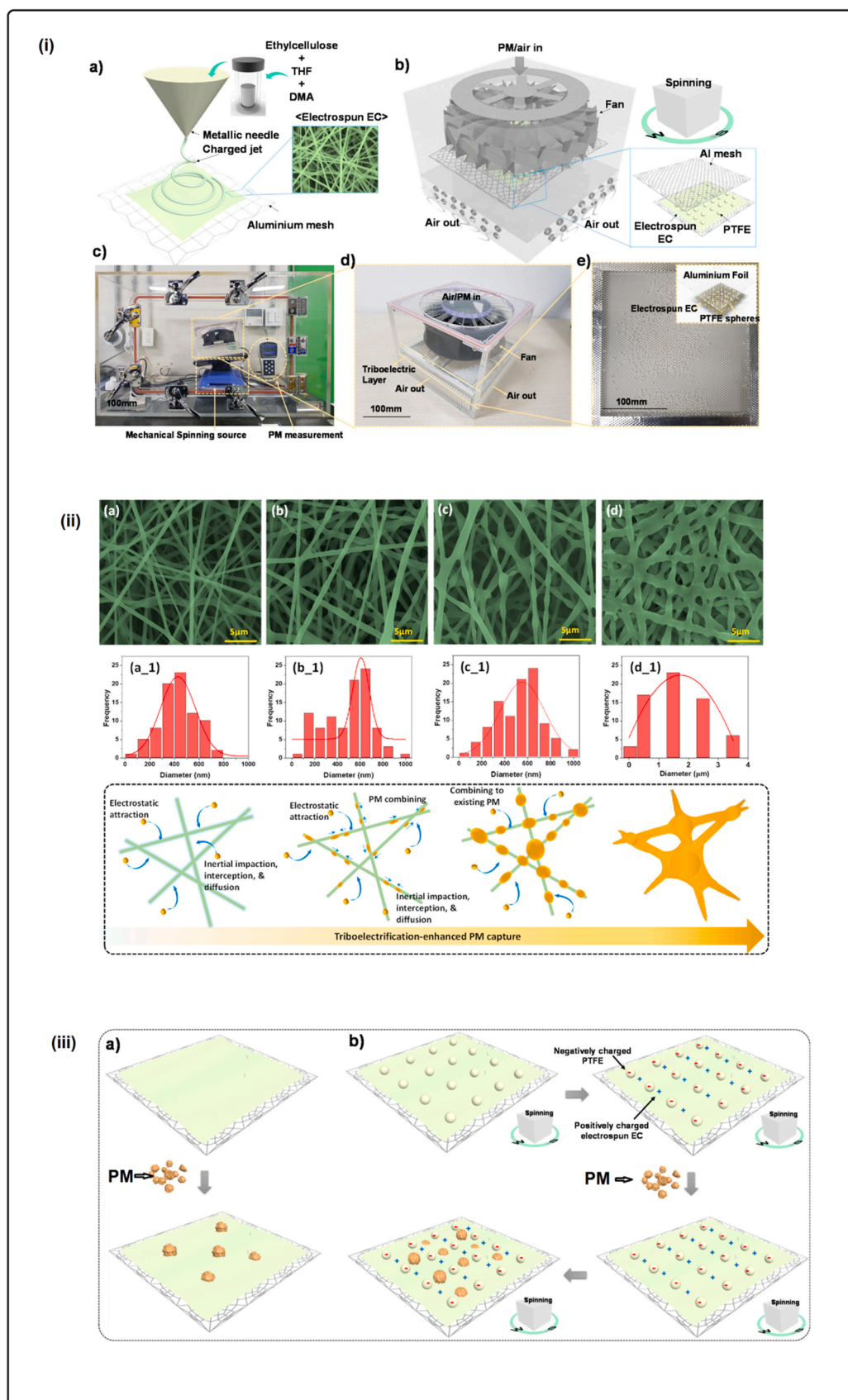


Fig. 13. (i) Conceptual presentation of (a) synthesis of electrospun EC over aluminium mesh utilising electrospinning technique, (b-e) lab-made triboelectric-based PM removal device, (ii) The PM capturing and the successive morphological changes of electrospun EC (a) before PM capture, (b-d) after PM capture with bulged morphology, (c) after PM capture with larger bulged morphology, (d) after PM capture with covered morphology, and their corresponding diameter distribution of nanofibers (a₁-d₁); and (iii) PM capture mechanism of triboelectric consisting of (a) electrospun EC and (b) electrospun EC and PTFE spheres (Cho et al., 2021). Reproduced with permission from Ref. Cho et al. (2021) Copyright 2021, Elsevier.

deposited MOFs and is a big challenge among researchers to select a particular technique keeping in view of its application.

Cellulose after functionalization with a suitable reagents, linkers, polymers or chemical may enhance the amount of deposition of MOFs or other bio-materials. However, this technique may increase the cost of resulting materials and thus may limit their large-scale production. For example, after carboxymethylation of cotton fabric (CCF), hydrophilicity of cotton fabric enhanced which in turn increased the deposited amount of MIL-100(Fe) (Lu et al., 2020). Also, polyvinyl pyrrolidone (PVP) coated CNFs lead to deposition of higher amount of various MOFs (HKUST-1, Al-MIL-53, Zn-MOF-74, ZIF-CO3-1) onto CNFs (S. Zhou et al., 2019b).

One more great challenge is to control properties of the cellulose based materials because of their heterogeneous nature (Matsumoto et al., 2019). More specifically BC, highly hydrated structure contains approx. 95% water and thus limits its utility in dry powder cases. It consists of randomly webbed CNFs of 20–30 nm diameters. Cellulose supports the processing and shaping of MOFs of other polymer matrix into a Fibers, foams, thin film, and 3D objects utilising 3D printing technology. The process generally requires gelatin or a mixture of metal salts and gelatine or solvents. These reagents enhance the mechanical toughness by simultaneously preserve the structure of the 3D printed materials intact. However, this technique will also increase the cost of the materials (Dou et al., 2020). Thus, investigations aiming to decrease these reagents are highly required.

In addition, combining surface modified polymers with electrospun nanofilters is a highly effective technique for creating high-performance green and sustainable nanofiber-based air filters. The electrospinning approach, without a doubt, improves the specific surface area of the filter as well as the overall performance of cellulose-based filters (Souzandeh et al., 2019). It should be noted, however, that research on cellulose-based air filtering technologies is still in its early stages. Researchers will also have a hurdle in synthesising at a much larger scale.

The chemical functionalization technique may enhance the cost of filter paper. So, it is a great challenge to develop a cost-effective filter paper. Another biggest challenge among researchers is to strike the right balance between flow resistance and filtration capability of filters. Since the filtration ability varies directly with mass and thickness of filters. So higher the thickness more will be filtration efficiency; however more thickness will also lead to enhanced flow resistance. So, while building complex filters, this difficulty should be completely addressed (Gough et al., 2021; Souzandeh et al., 2019). Blending multifunctional materials, such as cellulose or its derivatives, with other hydrophilic or hydrophobic fillers in single materials is another problem. Maintaining a balance between environmental stability and bio-deterioration is a difficult task because multifunctional materials are often very reactive towards atmospheric conditions.

Further, keeping in complex composition of air pollutants the future advanced air filters should address the following points. Ideal advanced air filter should be able to remove the various harmful pollutants such as VOCs, PM of various sizes and other toxic chemicals. Filters formed by stacking of layers of different materials, might not be suitable as stacking of layers will increase the thickness and ultimately the flow resistance.

The filters system having high filtration efficacy and low flow resistance are highly preferred. So, this should be fully addressed in future. The filters systems should have ability to sustain under varying atmospheric conditions such as at high humid conditions, high temperature, high contaminates concentration, etc. So, to develop the filter fulfilling the aforementioned factors should be a future challenge.

In this review article, we have reviewed the efforts made to utilize the cellulose or cellulose derivatives for developing of sustainable and bio-degradable filters. But to use the blending materials in proper ratio keeping in view life cycle, performance in varying conditions, biodegradable nature should be also a future plan.

The various challenges as described in aforementioned text may open new opportunities for the future studies on air-filtration materials

and air-filters. Further, for the development of air-filters, collective efforts from scientists working in multidisciplinary fields such as in biopolymer, nano bio-composites, materials processing and manufacturing, nanoscience and nanotechnology, particle physics and chemistry, bio-technology and others is highly required.

Utilizing cellulose-based membrane for CO₂ separation or adsorption is quite promising. The raw or functionalized cellulosic forms provide a very high selectivity for permeation of CO₂ over the CH₄ and N₂ or for adsorption of CO₂. However, no significant enhancement in gas permeability was noticed. However, membranes with ultrahigh CO₂/CH₄ selectivity and CO₂ permeance were created after combining it with MOFs or after functionalization with ILs (Nikolaeva et al., 2018; Raza et al., 2020). Nanocellulose-based nano filters have also been produced to degrade CO₂ or other VOCs after being combined with photo-catalyst, although most of these nano photo-catalyst are active in the UV region (Kwon et al., 2021; Lu et al., 2013). Thus, attempts to improve their photo catalytic activity and expand their involvement in the visible region are still needed.

Furthermore, efforts are still needed to understand the mechanism for gas transport in cellulose/MOFs membrane to offer a theoretical background for producing membrane having high CO₂ permeability and selectivity. Gas permeation through cellulose-based membrane is highly controlled by humid conditions so efforts should also make to develop membrane that could be utilised in wide range of humid conditions. Functionalization of cellulosic fibers may also improve their selectivity for gaseous molecules; careful consideration must be given to the type of surface tailoring technique to be used. Further, there is also urgent need to develop filters with adequate thickness for a sufficient gas flux.

6. Conclusion

We can see from the above explanation that cellulose-based filter systems have a lot of potential for air purification. Among the various techniques, the cellulose/protein concentrate based filter system (completely green ways) is the best since it does not require any hazardous chemicals, is fully sustainable, cost-effective, and has a high adsorption effectiveness against all three pollutants i.e. PM, VOCs and CO₂. However, if you want your cellulosic fibres to last longer or perform better in humid environments, hydrophobic treatment is a wonderful solution. In fact, all of the filters/membranes have performed admirably, and cellulose/MOF-based filter systems have received a lot of attention. The MOFs/cellulose filter/membrane system has demonstrated excellent contribution in PM, CO₂, and other VOCs removal. Other filter systems, on the other hand, have yet to be thoroughly investigated in terms of removing all three pollutants. The contribution of nano-photocatalyst or metal oxide-based filters/membrane to the degradation of VOCs or CO₂ is significant. However, little research has been done on its ability to remove PMs. Cellulose/CNTs or cellulose/polymer matrix filter systems are also accessible, however they may conceal us when we look at the expense of preparation. There are also cellulose/ILs-based filters, which are highly porous and could be a decent solution for environmentalists. All of the preceding material pertains to a filter system that can be employed as a mask or membrane in both indoor and outdoor situations. However, efforts have recently been undertaken to build a nanocellulose-based triboelectric system that can be put on a vehicle and purifies outdoor air on a big scale. Filter systems like these should be encouraged in the future to combat pollution issues.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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